

Testing Multidimensional Diffusion Processes for Stationarity

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Abstract

In this paper, we introduce a novel method for testing stationarity in multidimensional diffusion processes. Initially, we demonstrate that a stationary process exhibits identifiable behavior through the divergence of its occupation measure concerning its stationarity properties. We then employ a nonparametric estimator for the diffusion matrix, highlighting that its convergence rate is contingent on the stationary properties of the process. This is contrasted with a time domain estimator, which achieves a consistent convergence rate regardless of the process' stationarity. From this comparison, we derive a sequence of test statistics. Utilizing large deviation theory, we infer the likelihood of stationarity using the running maximum, which forms the foundation of our testing method. A simulation study revealed satisfactory performance in terms of empirical power and size, which we compared with established methods in the literature, including the Kwiatkowski-Phillips-Schmidt-Shin and Leybourne-McCabe tests. Furthermore, an application to two bivariate cryptocurrency portfolios provided additional insights into the empirical implications of our test. In particular, the obtained sequence of test statistics obeyed prior placed restrictions with regard to their autocovariance.

Key words: Nonparametric Stationarity Testing, Multidimensional Diffusion Processes, Stationary Processes, Hypothesis Testing.

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1 Introduction

Due to the substantial increase in the availability of financial data, often characterized by its dynamic nature, there has been significant interest from both the academic community as the financial industry in the modeling and forecasting of financial data. Simultaneously, the successful application of continuous time stochastic processes has found fertile ground in economics and finance, setting the foundations for modern econometric forecasting, usually under the assumption of stationarity. Yet, paradoxically, most of the theory devoted to testing for stationarity in these processes hinges on the assumption of discrete linear models from which a test statistic is usually obtained by means of unit-root testing, which has become obsolete in the realm of nonlinear continuous time processes. Although some progress has been made in constructing methods to test for stationarity in continuous time, most notably the methods in [Bandi and Corradi \(2014\)](#) and [Kanaya \(2011\)](#), a general method to infer the stationary properties for continuous time processes has yet to surface.

Generally, a strictly stationary process is defined to be a stochastic process with time invariant finite dimensional distributions. Despite its theoretical definition, in linear discrete models, inferring whether a process is stationary can simply be done by means of inference, where one tests for the invertibility of the lag polynomial ([Davidson \(2000\)](#)). Yet, contrary to the traditional linear discrete case, inferring the stationary characteristic of continuous time models, even when a parametric representation is available, is difficult. What is more is that, in many applications, assuming that a process has some parametric representation is often too restrictive, leading to misspecification. Nonetheless, inferring whether a process is stationary can still be done by means of its sample path properties. To achieve this, one can use the seminal results of [Darling and Kac \(1957\)](#) to distinguish stationary processes from their nonstationary counterparts. Namely, they were able to show that stationarity is a necessary condition for the occupation measure to diverge linearly to infinity, which crudely speaking is a measure on the fraction of time the process has spent in a certain section of its state space. Using the results of [Lazić and Sandrić \(2021\)](#) and [Lee and Trutnau \(2022\)](#), some mild regulatory conditions on the process consequently imply linear growth in the occupation measure to be a sufficient condition for stationarity, which implies that the behaviour of the occupation measure can (asymptotically) uniquely identify the stationary property. Given their success, a plethora of work was consequently devoted in obtaining estimates for the occupation measure, mainly by means of nonparametric estimation techniques, for example in [Bandi and Moloche \(2018\)](#) or [Puri \(1970\)](#). Therefore, this study aims to introduce a novel approach to testing for stationarity using the properties of the occupation measure. Given its nonparametric nature, this method will consequently be called a nonparametric stationarity test, following the terminology of [Bandi and Corradi \(2014\)](#) and [Kanaya \(2011\)](#).

In practice, to obtain a test statistic, one could in essence measure how fast the occupation measure diverges to infinity from which a conclusion can be drawn on account of whether the observed process was indeed stationary. Yet, a major drawback is that the exact distribution of the occupation measure is only known for a handful of processes, which makes inference hard. As a special case, [Bandi and Moloche \(2018\)](#) showed that the distribution, upon normalisation, depends on the stationarity of the process, and so if the *degree* of stationarity (see, e.g., [Meyn and Tweedie \(1993, 2012\)](#)) is known beforehand, then this approach would be viable. Yet, this defeats the purpose of stationarity testing and so rather than using the occupation measure directly, one needs to employ a methodology that solely relies on the divergence rates, without any reference to its weak (upon normalisation) limit.

An example comes in the form of the work by [Bandi and Corradi \(2014\)](#) who circumvented this by linearly normalising the occupation measure, which then converges to zero almost surely when a process is nonstationary while a stationary process results in a nontrivial random variable. Consequently, they inferred the nonstationarity using a randomised test, solely based on observing a single trajectory (or realisation) of the process. However, a significant drawback in their approach is the necessity to calculate the test statistic for an extensive class of functions. This was mitigated by computing the test statistic for a strategically selected subset. Nevertheless, this simplification inherently results in a trade-off with the test's statistical power. Another approach was introduced in [Kanaya \(2011\)](#), wherein the authors argued that the stationarity property originates from the characteristics of the generator of the Markov process. By analyzing the behavior of this generator, they inferred the stationary characteristic. Namely, a stationary sequence, by definition, has a distribution invariant to shifts in time, implying that, in theory, the generator should be zero for all test functions within its domain. However, similar to the approach in [Bandi and Corradi \(2014\)](#), this method depends on the validity of an assertion for a wide range of possible configurations. Consequently, any reduction in this set results in a corresponding loss of power.

Therefore, we take a different approach to test for stationarity. Namely, we will use an approach that resembles the Durbin-Wu-Hausman test, which asserts whether a given statement is plausible on account of the behaviour of two estimators. In this framework, this amounts to comparing two estimators for the diffusion coefficient of the process. First, we take the time domain estimator of [Jacod and Protter \(2011\)](#), which is consistent and has the central limit property provided that certain regulatory conditions are met. Second, we use the state domain estimator of [Bandi and Moloche \(2018\)](#), which is also consistent and has the central limit property, however, unlike the estimator of [Jacod and Protter \(2011\)](#), the speed at which the state domain estimator is asymptotically normal depends on whether the process is stationary or not. Consequently, by the behaviour of the occupation measure, the

state domain smoother is inherently *more* consistent if the process is stationary, as the available data has an asymptotic linear growth rate.

However, one of the fundamental difficulties in this approach is that it results in a sequence of test statistics rather than a single random variable, as a consequence to the pairwise comparison of the estimators. Although several methods exist for multiple hypothesis testing, most notably the Benjamini-Hochberg procedure introduced in [Benjamini and Hochberg \(1995\)](#) or the Benjamini-Yekutieli approach of [Benjamini and Yekutieli \(2001\)](#), these do not take the explicit Gaussian structure of the obtained sequence in consideration. Motivated by the results of [Berman \(1964, 1982\)](#) and [Pickands \(1969\)](#), an alternative approach is available, which uses the distribution of the running maximum to assert the likelihood of stationarity on account of the obtained individual test statistics. Therefore, the contributions of this paper are twofold: first, we introduce a new method to identify the stationary characteristic of diffusion processes under minimal assumptions and secondly, we introduce the method of the running maximum to obtain a distribution of the global test statistic.

1.1 Nonparametric Stationarity Testing

Hoping to avoid parametric assumptions, one might resort to nonparametric tests for stationarity, which essentially involves nonparametric techniques to infer whether a process is stationary. Before introducing the existing literature in [Section 2](#), we will briefly discuss the intuition behind the nonparametric approach in this section.

That is, consider a d -dimensional stochastic process $X = (X_t)_{t \geq 0}$ with state space $\mathbb{R}^d$ serving as the strong solution to the time-homogeneous Itô stochastic differential equation $dX_t = b(X_t)dt + \sigma(X_t)dW_t$, where b, σ are Borel measurable functions whose properties are laid out in [Section 2](#) and $W = (W_t)_{t \geq 0}$ is a d -dimensional Brownian motion. In most applications, these types of processes are observed discretely on a finite interval $[0, T]$ with sampling frequency $\Delta_n = T/n$ consisting of n observations. To this end, we will follow [Bandi and Moloche \(2018\)](#) to define the empirical occupation measure $L_{n,T}(x) = \Delta_n \sum_{i=1}^n \mathbb{1}\{|X_{i\Delta_n} - x| \leq h_n\}$ for a bandwidth sequence h_n and any $x \in \mathbb{R}^d$. Standard results in nonparametric estimation, for example in [Puri \(1970\)](#) or [Dudley \(2014\)](#), consequently dictate that we can obtain the central limit result of the pointwise estimator $\hat{c}(x)$ of $c(x) = \sigma(x)\sigma(x)^\top$, provided various regulatory conditions are met. In fact, theorem 6 in [Bandi and Moloche \(2018\)](#) readily implies that

$$\sqrt{L_{n,T}(x)h_n^d/\Delta_n} \text{vech}(\hat{c}(x) - c(x) - \Gamma(x)) \xrightarrow{d} \mathcal{N}(0, \Sigma) \quad (1.1.1)$$

as $n, T \rightarrow \infty$, $h_n \rightarrow 0$ and $\Delta_n \rightarrow 0$ where Σ is some covariance matrix, $\Gamma(x)$ a matrix of biases and $\mathcal{N}$ denotes the normal distribution.

Interestingly, [Gettoor \(1980\)](#) and [Meyn and Tweedie \(2012\)](#) showed that if X is a stationary process, then $L_{n,T}(x) \sim T$ for every x , meaning that for two positive constants $C_1(x)$ and $C_2(x)$, $C_1(x)T \leq L_{n,T}(x) \leq C_2(x)T$ for sufficiently large T . What is more is that [Meyn and Tweedie \(1993, 2012\)](#) argue that a nonstationary process satisfies $L_{n,T}(x) = o_p(T)$ as $n, T \rightarrow \infty$. Given that $\Delta_n = T/n$, fixing the rate of convergence in Equation (1.1.1) to correspond to the stationary case essentially implies that

$$\sqrt{nh_n^d} \text{vech}(\hat{c}(x) - c(x)) = \sqrt{\frac{T}{L_{n,T}(x)}} \sqrt{L_{n,T}(x)h_n^d/\Delta_n} \text{vech}(\hat{c}(x) - c(x)). \quad (1.1.2)$$

Hence, in the stationary case, $L_{n,T}(x) \sim T$ and Equation (1.1.2) has a (finite) weak limit. Conversely, a nonstationary process has $L_{n,T}(x) = o_p(T)$ as $n, T \rightarrow \infty$, which means that Equation (1.1.2) will diverge to plus or minus infinity. As a consequence, the asymptotic behaviour of Equation (1.1.2) is tied to the stationary properties of the underlying process.

However, it is generally unknown what $c(x)$ is at a given point $x \in \mathbb{R}^d$. This can be solved by replacing $c(x)$ with an estimator $\hat{c}'$, independent to $\hat{c}$ and the stationary properties of X . This comes in the form of the so-called time domain smoother, for example the estimators of [Jacod and Protter \(2011\)](#), [Fan and Wang \(2008\)](#) and [Fan et al. \(2007\)](#), which are methods that use information in the vicinity of a time point, rather than a state oriented approach as in [Bandi and Moloche \(2018\)](#). That is to say, if $\hat{c}'$ is another estimator of the diffusion coefficient independent to $\hat{c}$ satisfying $\sqrt{k_n} \text{vech}(\hat{c}'(x) - c(x)) \xrightarrow{d} \mathcal{N}(0, \Sigma')$, then granted that $nh_n^d/k_n \rightarrow C > 0$, we obtain

$$\sqrt{nh_n^d} \text{vech}(\hat{c}(x) - \hat{c}'(x)) \xrightarrow{d} \mathcal{N}(0, \Sigma + C\Sigma'), \quad (1.1.3)$$

if and only if X is stationary, where k_n denotes the number of data points used by $\hat{c}'$. Repeating Equation (1.1.3) and normalising each instance by the positive square root of $\Sigma + C\Sigma'$ consequently yields a sequence of standard normal random variables from which a test statistic can be constructed.

1.2 Outline

We now give a brief overview of the outline of this paper: In Section 2, we will provide the theoretical framework on which the test is constructed along with a detailed construction of the processes and an explanation of the required assumptions. In particular, we will introduce the underlying theory of the two estimators, after which we will show that the two methods are independent. Next, in Section 3, we will construct the test statistic using the results of Section 2. We start by explaining the difficulties of multiple hypothesis testing where we give an overview of some existing literature followed by Section 3.1, which introduces the running maximum test statistic, complemented by various remarks and shortcomings. In Section 4, we

will investigate the properties of the proposed test by means of simulation, which will pave way for Section 5, where we apply the method to two bivariate crypto portfolio's to assert whether they are stationary. Additionally, in Section 5, we will discuss various practical difficulties regarding the test, and how these can, to an extent, be reduced using certain configurations. We close off with Section 6, where we draw final remarks on the results and we provide various ideas on further research. Given the complexity of the underlying theory regarding the stationary characteristics of continuous time models, in Appendix A, a thorough list of supplementary results is provided.

2 Theoretical Framework

In this section, we will introduce the theory of stationarity testing in continuous time. We start by introducing the processes along with their assumptions, each supplemented with various remarks on their interpretation and implications. Additionally, we discuss the results by Bandi and Moloche (2018) and Jacod and Protter (2011) with respect to our given framework. We close the chapter by showing that, in an asymptotic sense, the above two methods are independent under stationarity.

2.1 Preliminary Assumptions

Indeed, as discussed in the introduction, we aim to construct a nonparametric test statistic resembling the Durbin-Wu-Hausman method. To do so, we will employ the state domain estimator of Bandi and Moloche (2018) and the time domain smoother in Jacod and Protter (2011). Given that both methods rely on some mild regulatory conditions, we start this chapter by placing the assumptions needed to define the test statistic later on.

Remark 1 (Notation). For functions $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$, we will use the notions of Stroock and Varadhan (1997) to define $\|f(x)\|$ to be the Euclidean norm on $\mathbb{R}^k$ while $\|f\|_p = (\int |f|^p dx)^{1/p}$ denotes the L^p -(semi)norm with respect to the Lebesgue measure unless stated otherwise. If $f(x)$ is matrix valued, then $\|f(x)\| = \sum_{ij} f_{ij}(x)^2$ denotes the Hilbert-Schmidt norm. Additionally, we will write $\langle x, y \rangle = \sum_{i=1}^d x_i y_i$ to denote the inner product for any two vectors $x, y \in \mathbb{R}^d$.

Assumption 1. The process X is a continuous d -dimensional diffusion process in $\mathbb{R}^d$, defined as the solution

$$X_t = X_0 + \int_0^t b(X_s) ds + \int_0^t \sigma(X_s) dW_s \quad \forall t \geq 0$$

where $W = (W_t)_{t \geq 0}$ is a d -dimensional Brownian motion and b, σ are Borel measurable functions satisfying the Lipschitz and linear growth conditions. That is, for two finite positive constants C_1, C_2

$$\|b(x) - b(y)\| + \|\sigma(x) - \sigma(y)\| \leq C_1 \|x - y\|$$

for every $x, y \in \mathbb{R}^d$ and

$$\|b(x)\| + \|\sigma(x)\| \leq C_2(1 + \|x\|).$$

Remark 2. Assumption 1 simply states that we are working with an Itô diffusion process where the Lipschitz and linear growth conditions are placed to ensure that a solution exists, and moreover that it is unique in the sense that any other solution Y must then be indistinguishable from X . Here, following the standard interpretation of Revuz and Yor (1999), the continuity condition is understood as almost sure continuity in the sense that the set $\{\omega : X(\omega) \in C[0, \infty)\}$ contains a set of probability one. Lastly, these conditions are placed so that the process does not explode in finite time, details are to be found in section 10.2 in Stroock and Varadhan (1997).

Assumption 2. There exists a constant $\gamma > 1$ such that

$$\gamma^{-1}\|y\| \leq \langle y, c(x)y \rangle \leq \gamma\|y\|$$

for every vector $y \in \mathbb{R}^d$ and $x \in \mathbb{R}^d$ where $c(x) = \sigma(x)\sigma(x)^\top$ denotes the diffusion matrix.

Remark 3. Assumption 2 is the so-called uniform ellipticity assumption. What we will see is that for a process to be stationary, one of the conditions is that no cyclic behaviour can exist in any of its sample paths. Namely, as was shown in Meyn and Tweedie (1993), a necessary and sufficient condition for a process to be ergodic is that it is aperiodic and positive Harris recurrent. From Lazić and Sandrić (2021), we know that Assumption 2 naturally implies aperiodicity, while the linear growth of the occupation measure implies the latter. What is more is that, by Bogachev et al. (2012), Assumption 2 implies that any invariant measure is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^d$, and the Radon-Nikodym derivative is everywhere positive, which is a constraint needed in order to employ the results of Bandi and Moloche (2018).

Remark 4. On a more technical note, Lazić and Sandrić (2021) and Prohaska (2021) argued that Assumption 2 is placed to ensure that the associated Markov semigroup, as defined in Equation (A.1.1), converges in total variation norm to its invariant measure. The reason as to why this is so important is that, unlike the method of unit-root testing, our approach relies on the equivalence between divergence rates of occupation measures and stationarity. As mentioned by Eisner et al. (2015), Gettoor (1980) and Shin and Trutnau (2017), if Assumption 2 is satisfied, then linear divergence of occupation measures implies convergence of the semigroup in total variation norm to an invariant finite measure, which can then be normalised to a probability measure. This by definition corresponds to the phenomenon of ergodicity and so the stationarity property comes as a by-product, conditional on initialisation on the invariant measure. Indeed, if we simply drop Assumption 2, then there exists examples in which the topology induced by the total variation norm becomes

too rough in order for the semigroup to converge to its invariant measure, see for example [Lazić and Sandrić \(2021\)](#) and [Sandrić \(2017\)](#) and the references therein. Interestingly enough, this convergence might still take place in weaker metrics, such as the Wasserstein distance. A complete proof of the relation between stationarity, occupation measures, ergodicity and semigroups is the topic of [Appendix A.1](#).

Assumption 3. *The process is Harris recurrent.*

Remark 5. *Assumption 3 is a condition that ensures that all sets of positive measure are revisited by the process infinitely often, which is a necessary condition to allow the state domain estimator to converge. More importantly, this assumption ensures that there exists an invariant measure ϕ for the process due to [Gettoor \(1980\)](#), which is defined to be a measure such that*

$$\phi(A) = \int_{\mathbb{R}^d} \mathbb{P}_x(X_t \in A) d\phi(x)$$

for every $t \geq 0$ and every $A \in \mathcal{B}(\mathbb{R}^d)$, see for example [Meyn and Tweedie \(1993\)](#).

Remark 6. *Intuitively, Assumption 3 forces the occupation measure to diverge to infinity with probability one, which is needed in order to employ the results of [Darling and Kac \(1957\)](#) and [Bandi and Moloche \(2018\)](#). Although this assumption is satisfied in many real-world processes, it naturally restricts any usage of processes which have a diverging drift. Examples include the Brownian motion with drift $W_t^{\text{drift}} = W_t + bt$ for some $|b| > 0$ or processes which remain in a set B of their state space indefinitely the moment the path hits B .*

The assumptions above are flexible and allow for a broad range of diffusion processes. In fact, Assumption 1 and Assumption 3 are common assumptions in literature while Assumption 2 is a minor regulatory constraint. [Lazić and Sandrić \(2021\)](#) argued that, in many applications, it is generally known beforehand whether a process shows cyclic behaviour or not, and so assuming this additional condition should not cause for any problems.

2.2 The Time Domain Smoother

We start with explaining the asymptotic properties of the time domain smoother of [Jacod and Protter \(2011\)](#). As the name suggests, the method is based on information in the vicinity of a fixed point in time. To give a statement of the result, we first need to introduce some notation to align with the work of [Jacod and Protter \(2011\)](#). Given that in practice all processes are discretely observed, we will assume that our process X is discretely observed on an interval $[0, T]$ with sampling frequency Δ_n , where n will denote the total number of observations, which then implies that $\Delta_n = T/n$. Now it is well known, for example by arguments in [Bandi and Moloche \(2018\)](#) or [Revuz and Yor \(1999\)](#), that the drift and diffusion coefficient can be represented by the first and second infinitesimal moment, meaning that

$$\mathbb{E}_x(X_t^i - x^i) = tb_i(x) + o(t), \quad \mathbb{E}_x(X_t^i - x^i)(X_t^j - x^j) = tc_{ij}(x) + o(t) \quad (2.2.1)$$

as $t \downarrow 0$. It is this characteristic on which the time domain estimator is constructed. Namely, if $\Delta_n^i X_t = X_{t+i\Delta_n} - X_{t+(i-1)\Delta_n}$, and if k_n is a sequence so that $k_n \rightarrow \infty$ but $k_n^2 \Delta_n \rightarrow 0$, and

$$\hat{c}_{\text{TD},t}^n = \frac{1}{\Delta_n k_n} \sum_{m=1}^{k_n} \Delta_n^m X_t \Delta_n^m X_t^\top$$

denotes the time domain estimator, then it is reasonable to assume that $\hat{c}_{\text{TD},t}^n \rightarrow c_t$ in probability by virtue of Equation (2.2.1). This turns out to be true, as $\hat{c}_{\text{TD},t}^n \xrightarrow{p} c_t$ by theorem 9.3.2 in [Jacod and Protter \(2011\)](#). In fact, this result can be strengthened to a central limit result:

Theorem 2.1 (Jacod-Protter (2011)). *Under assumptions 1 to 3, for any $t \geq 0$ as $n, k_n \rightarrow \infty$ so that $k_n^2 \Delta_n \rightarrow 0$*

$$\sqrt{k_n} \text{vech}(\hat{c}_{\text{TD},t}^n - c_t) \xrightarrow{d} Z$$

where Z is a normally distributed random vector with mean zero and covariance matrix

$$\Sigma_{\text{TD},t} = \text{diag vech}(V) P_D (c_t \otimes c_t) P_D^\top$$

where $\otimes$ is the Kronecker product, $P_D = (D^\top D)^{-1} D^\top$ with $D \in \mathbb{R}^{d^2 \times d(d+1)/2}$ the standard duplication matrix and V is a matrix with entries

$$V_{ij} = \begin{cases} 2 & \text{if } i = j \\ 1 & \text{else} \end{cases}.$$

Remark 7. Here, contrary to the original statement of [Jacod and Protter \(2011\)](#), we express the weak limit in terms of the half-vectorised variant, rather than to obtain a weak limit for the matrix process, which is easily obtained by simple matrix manipulation similar to [Davidson \(2000\)](#), drastically reducing the dimensions of the resulting limiting random variable. Additionally, the original result in [Jacod and Protter \(2011\)](#) is stated in terms of stable convergence. This however should not cause for issues as it is well known, for example due to [van der Vaart and Wellner \(2013\)](#), that stable convergence implies convergence in distribution.

Remark 8. The condition that $k_n^2 \Delta_n \rightarrow 0$ can be weakened to $k_n^2 \Delta_n \rightarrow \beta$, where β is a nonnegative scalar. Indeed setting $\beta = 0$ yields the result of Theorem 2.1, yet letting $\beta \in (0, \infty)$, which essentially changes the joint convergence speeds of k_n and Δ_n as functions of n , corresponds to a different asymptotic limit. In fact, if $\beta > 0$, the structure of Z involves additional components not formulated in Theorem 2.1. What is more is that in the extreme case $\beta = \infty$, it is still possible to obtain a central limit result. It would be interesting to assert if the convergence rate of the stationarity test, to be introduced below, can be improved using an informed choice to β . This however is beyond the scope of this paper and is left for future research, the interested reader is referred to chapter 13 in [Jacod and Protter \(2011\)](#) (and in particular theorem 13.3.3).

A remarkable feature of the limiting distribution in Theorem 2.1 is that it does not depend on the stationarity of the underlying process. In fact, it can even be shown that the estimator works for a much larger class of processes, called Itô jump semimartingales. Namely, by some simple adjustments to the estimator, it can be shown that the jumps can simply be truncated out of the estimator, and so if some minor regulatory constraints on the jumps are placed, then the same asymptotic properties should arise. The interested reader is referred to Jacod and Protter (2011) and Lazić and Sandrić (2021).

2.3 The State Domain Estimator

Next, we will focus on estimating the diffusion matrix by virtue of the state domain estimator. Given that we will be working with kernel regression algorithms, we require some additional assumptions with regard to the kernel function K and its associated bandwidth sequence $h_{n,T}$ which is a function of n and T .

Assumption 4. *The kernel function K is a product kernel $K(x) = \prod_{i=1}^d k(x^i)$ where $k(x)$ is a nonnegative bounded continuous symmetric function on $\mathbb{R}$ satisfying $\int k(x)dx = 1$, $\int k^2(x)dx < \infty$ and $\int x^2 k(x)dx < \infty$. Additionally, we assume that there exists a nonnegative function $D(y, \varepsilon)$ that satisfies*

$$|K(x) - K(y)| \leq D(y, \varepsilon) \|x - y\|$$

for every $x, y \in \mathbb{R}^d$ so that $\|x - y\| \leq \varepsilon$. Moreover

$$\lim_{\varepsilon \rightarrow 0} \int D(x, \varepsilon) dx < \infty \quad (2.3.1)$$

and, for any $y < \infty$

$$\int D(x, y) d\phi(x) < \infty \quad (2.3.2)$$

where ϕ denotes the invariant measure of the process.

Remark 9. *Contrary to standard results in kernel regression, Assumption 4 additionally assumes the seemingly unrelated restrictions of Equations (2.3.1) and (2.3.2). However, Bandi and Moloche (2018) mentions that these restrictions guarantee that the kernel function is in essence sufficiently regular in order to obtain the desired convergence in distribution.*

Using the kernel K as in Assumption 4, we can define the Nadaraya-Watson estimator for the volatility matrix as

$$\hat{c}_{SD}^n(x) = \frac{1}{\Delta_n} \frac{\sum_{j=1}^n K_{h_{n,T}}(X_{t_j} - x)(X_{t_j} - X_{t_{j-1}})(X_{t_j} - X_{t_{j-1}})^\top}{\sum_{j=1}^n K_{h_{n,T}}(X_{t_j} - x)}$$

where we write $K_{h_{n,T}}(x) = \frac{1}{h_{n,T}^d} K(x/h_{n,T})$ to emphasize that the kernel depends on a bandwidth sequence converging as $n, T \rightarrow \infty$. Here, we make a minor adjustment on

the empirical occupation measure, by redefining $L_{n,T}(x) = \Delta_n \sum_{j=1}^n K_{h_{n,T}}(X_{t_j} - x)$, i.e., it is a weighted sum of all observations based on their distance to $x \in \mathbb{R}^d$, normalised by $\Delta_n = T/n$ measured with respect to the kernel K .

As briefly mentioned in Assumption 2, the state domain estimator relies on the invariant density. Indeed, in many applications of nonparametric statistics, one needs to correct for the asymptotic bias of the estimator. Although the results in Bandi and Moloche (2018) account for this discrepancy, they do so by means of the measure $\tilde{\phi}(x) = \phi(dx)/dx$, existence of which is guaranteed by Bogachev et al. (2012) granted Assumption 2 is met.

Theorem 2.2 (Bandi-Moloche (2018)). *Assume that $L_{n,T}(x) \rightarrow \infty$ and*

$$\frac{h_{n,T}^d L_{n,T}(x)}{\Delta_n} \xrightarrow{p} \infty, \quad \frac{h_{n,T}^{d+4} L_{n,T}(x)}{\Delta_n} \xrightarrow{p} C, \quad h_{n,T}^{d-4} L_{n,T}(x) \Delta_n \xrightarrow{p} 0$$

as $n, T \rightarrow \infty$ with $\Delta_n \rightarrow 0$ for a positive constant C , then

$$\sqrt{\frac{h_{n,T}^d L_{n,T}(x)}{\Delta_n}} \text{vech}(\hat{c}_{\text{SD}}^n(x) - c(x) - \Gamma(x)) \xrightarrow{d} \Sigma_{\text{SD}}^{1/2}(x) \mathcal{N}\left(0, \left(\int k^2(u) du\right)^d I_{d(d+1)/2}\right)$$

where I_m is the m -dimensional identity matrix,

$$\Sigma_{\text{SD}}(x) = P_D(2c(x) \otimes c(x))P_D^\top$$

and $\Gamma(x)$ is a $d \times d$ matrix with elements

$$\Gamma_{ij}(x) = h_{n,T}^2 \left(\int u^2 k(u) du \right) \left(\sum_{k=1}^d \frac{\partial c_{ij}(x)}{\partial x^k} \frac{\frac{\partial \tilde{\phi}(x)}{\partial x^k}}{\tilde{\phi}(x)} + \frac{1}{2} \sum_{k=1}^d \frac{\partial^2 c_{ij}(x)}{\partial x^k \partial x^k} \right).$$

Corollary 2.1. *If, in addition to the assumptions of Theorem 2.2, the process X is stationary, then*

$$\sqrt{nh_{n,T}^d} \text{vech}(\hat{c}_{\text{SD}}^n(x) - c(x) - \Gamma(x)) \xrightarrow{d} \frac{\Sigma_{\text{SD}}^{1/2}(x)}{\sqrt{p(x)}} \mathcal{N}\left(0, \left(\int k^2(u) du\right)^d I_{d(d+1)/2}\right)$$

where $p(x) = \tilde{\phi}(x)/\tilde{\phi}(\mathbb{R}^d)$ denotes the normalised invariant measure.

A proof of Theorem 2.2 can be found in Bandi and Moloche (2018), while the result of Corollary 2.1 is a direct consequence to remarks 8 and 12 in Bandi and Moloche (2018). Here, as mentioned above, we find that the speed at which $\hat{c}_{\text{SD}}^n(x)$ achieves the central limit property is now dependent on $L_{n,T}(x)$ and hence the degree of stationarity of the underlying process. This remarkable result is far from trivial and is actually due to the seminal work of Darling and Kac (1957) along with various extensions in Kasahara (1975). Namely, for the class of Harris recurrent Markov processes, if there exists a nonnegative function f such that

$$\lim_{T \rightarrow \infty} \frac{1}{U(T)T^\alpha} \mathbb{E}_x \int_0^\infty \exp\left(-\frac{t}{T}\right) f(X_t) dt = C_X$$

for a positive process specific constant C_X where $U(T)$ is a (Karamata) slowly varying function, i.e., $\lim_{n \rightarrow \infty} U(kn)/U(n) = 1$ for any $k > 0$, then

$$\lim_{T \rightarrow \infty} \mathbb{P}_x \left(\frac{1}{C_X U(T) T^\alpha} \int_0^T f(X_t) dt \leq x \right) = G_\alpha(x)$$

where $G_\alpha(x)$ is the cumulative Mittag-Leffler distribution with parameter $\alpha \in [0, 1]$ and $\mathbb{P}_x(\cdot) = \mathbb{P}(\cdot \mid X_0 \sim x)$ denotes the dependency on initialisation.

Two remarks are in order. First and foremost, the function U seems rather arbitrary, given that [Darling and Kac \(1957\)](#) simply assume that $U(T)$ exists without further clarification on any existence conditions. Fortunately, [Kasahara \(1975\)](#) showed that under assumptions 1 and 3, the diffusion process admits a Karamata function. Second, we see that α is the scalar coefficient which determines the rate of divergence to infinity, which will inevitably determine whether a process is stationary, and is called the regularity index of X by [Meyn and Tweedie \(2012\)](#). Some caution is needed to link the behaviour of $U(T)T^\alpha$ to the stationary characteristics of the underlying process. Namely, a stationary process will be characterised by a regularity index $\alpha = 1$ with $U(T) = 1$ as a consequence to [Revuz and Yor \(1999\)](#), yet a process with regularity index 1 is not necessarily stationary, but rather positive Harris recurrent. However, in case that X is initialised at the invariant measure and Assumption 2 is met, then positive Harris recurrence implies stationarity and vice versa. Moreover, using theorem 3 in [Bandi and Moloche \(2018\)](#), the function $T \mapsto U(T)T^\alpha$ can be replaced by the empirical occupation measure $L_{n,T}(x)$ to obtain a similar result. An overview of the necessary and sufficient conditions for stationarity with respect to the divergence rates of $L_{n,T}$ is the topic of Appendix A.1 with the final claim formulated in Theorem A.2.

Remark 10. *To give a better understanding of the results by [Darling and Kac \(1957\)](#), we will give some examples and explain how their behaviour will help in constructing a test. Take for example a scalar Brownian motion $W = (W_t)_{t \geq 0}$, from versions of the Kallianpur-Robbins Theorem (c.f. remark 4 in [Bandi and Moloche \(2018\)](#) and theorem 1 in [Darling and Kac \(1957\)](#)), it holds that*

$$\lim_{T \rightarrow \infty} \mathbb{P}_x \left(\frac{1}{C_X \sqrt{T}} \int_0^T \mathbb{1}_B(W_t) dt \leq u \right) = G_{\frac{1}{2}}(u) = \frac{1}{\sqrt{\pi}} \int_0^u e^{-x^2/4} dx, \quad \forall u \geq 0.$$

What this means is that the so-called regularity index (or α -index) is $1/2$, which is sublinear. This is expected given that a Brownian motion is recurrent yet not stationary nor ergodic. In fact, a scalar Brownian motion converges to the Lebesgue measure in total variation norm, which consequently serves as its invariant measure. Hence, if we where to estimate the diffusion coefficient of a scalar Brownian motion by means of the estimator of [Bandi and Moloche \(2018\)](#), which is simply $c(x) = 1$ for every $x \in \mathbb{R}$, then $\sqrt{nh_{n,T}^d}/\sqrt{T}(\hat{c}(x) - 1) \xrightarrow{d} \mathcal{N}(0, \Sigma)$. This is fundamentally different from stationary unit variance processes, from which the Darling-Kac theorem of [Darling and Kac \(1957\)](#) ensures that $\sqrt{nh_{n,T}^d}(\hat{c}(x) - 1) \xrightarrow{d} \mathcal{N}(0, \Sigma)$.

Another interesting consequence is that nonstationary processes suffer from the double curse of dimensionality, one in terms of their occupation measure and the other in terms of the number of parameters. A good example is to consider a planar Brownian motion W , simply defined to be a two-dimensional vector of independent Brownian motions, by which [Kasahara \(1975\)](#) demonstrated that

$$\lim_{T \rightarrow \infty} \mathbb{P}_x \left(\frac{1}{C_X \log T} \int_0^T \mathbb{1}_B(W_t) dt \leq u \right) = G_0(u) = 1 - e^{-u}, \quad \forall u \geq 0.$$

Here, we see that increasing the dimension from one to two decreases the speed from $\sqrt{T}$ to $\log T$ with regularity index $\alpha = 0$, which is rather extreme given a unit increase in dimension. What is more is that a Brownian motion of dimension $d \geq 3$ is known not to be recurrent as a consequence to Polya's recurrence theorem.

As we will employ the discrepancy in divergence rates between stationary and nonstationary processes in terms of the [Bandi and Moloche \(2018\)](#) state domain smoother, a requirement will inevitably be that a nonstationary process must satisfy $\sqrt{nh_{n,T}^d} \text{vech}(\hat{c}_{\text{SD}}^n(x) - c(x) - \Gamma(x)) \xrightarrow{\text{a.s.}} \pm\infty$. That is to say, a nonstationary process almost surely diverges when the convergence rate of a stationary process is exercised.

Proposition 2.1. *Let X be a nonstationary process satisfying the conditions in Assumptions 1 to 3 and assume that the kernel K satisfies Assumption 4, then*

$$\sqrt{nh_{n,T}^d} \text{vech}(\hat{c}_{\text{SD}}^n(x) - c(x) - \Gamma(x)) \rightarrow \pm\infty$$

almost surely for every $x \in \mathbb{R}^d$.

Proof. First and foremost, observe that as $\Delta_n = T/n$, then

$$nh_{n,T}^d = \frac{T}{L_{n,T}(x)} \frac{nh_{n,T}^d L_{n,T}(x)}{T} = \frac{T}{L_{n,T}(x)} \frac{h_{n,T}^d L_{n,T}(x)}{\Delta_n}. \quad (2.3.3)$$

But Theorem 2.2 ensures that $\sqrt{\frac{h_{n,T}^d L_{n,T}(x)}{\Delta_n}} \text{vech}(\hat{c}_{\text{SD}}^n(x) - c(x) - \Gamma(x)) \xrightarrow{d} \mathcal{N}(0, \Sigma(x))$ for some $\Sigma(x)$ finite. Yet, the results of Appendix A.1, and in particular Theorem A.1 and Proposition A.1 imply that $L_{n,T}(x) = o_p(T)$ as $n, T \rightarrow \infty$ and so $T/L_{n,T}(x) \rightarrow \infty$ for every $x \in \mathbb{R}^d$ in case that X is nonstationary. To obtain the almost sure convergence, we can simply use the second Borel-Cantelli lemma along with tail estimates for normal distributions to obtain the desired result. That is, by the Portmanteau lemma and Theorem 2.2, for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that whenever $n > N$, then

$$\mathbb{E}|f\left(\sqrt{L_{n,T}(x)h_{n,T}^d/\Delta_n} \text{vech}(\hat{c}_{\text{SD}}^n(x) - c(x) - \Gamma(x))\right) - f(\mathcal{N}(0, \Sigma))| \leq \varepsilon$$

for every continuous bounded function f on $\mathbb{R}^{d(d+1)/2}$. Hence, for any $z \in \mathbb{R}^{d(d+1)/2}$ and $n > N$, we can write

$$\mathbb{P}(Z \leq z) - 2\varepsilon \leq \mathbb{P}\left(\sqrt{L_{n,T}(x)h_{n,T}^d/\Delta_n} \text{vech}(\hat{c}_{\text{SD}}^n(x) - c(x) - \Gamma(x)) \leq z\right) \leq \mathbb{P}(Z \leq z) + 2\varepsilon \quad (2.3.4)$$

for $Z \sim \mathcal{N}(0, \Sigma(x))$. Now if $(\xi_n)_{n \geq 1}$ is an independent sequence of centered normal random variables with variance $\sigma_n^2 \rightarrow \infty$ for a strictly increasing sequence $(\sigma_n^2)_{n \geq 1}$, then for any $z > 0$ finite

$$\sum_{n=1}^{\infty} \mathbb{P}(|\xi_n| \geq z) \geq \sum_{n=1}^{\infty} \frac{c_1 z}{z^2 + 1} e^{-x^2/2\sigma_n^2} = \infty \quad (2.3.5)$$

by Mill's ratio. Hence $\mathbb{P}(\limsup_{n \rightarrow \infty} |\xi_n| \geq z) = 1$ by the second Borel-Cantelli lemma for every z . Applying Equation (2.3.5) to Equation (2.3.4) along with the squeeze theorem yields the desired conclusion. $\square$

2.4 Independence Properties

In this section, we will show that the two estimators above are in fact independent provided that the underlying process is stationary. This can in essence be done in two steps. First, much like the approach in [Fan et al. \(2007\)](#), we truncate the state domain estimator to exclude data in an increasingly larger ball centered around t , after which we will argue that this then implies independence. Second, we will show that, if the cardinality of the ball is negligible with respect to the total number of data points, then the truncated estimator is equal to the original (in probability) as $n, T \rightarrow \infty$.

Before we state the theorem, let us first explore the relation between the number of data needed in the state domain estimator with respect to the time domain estimator. Namely, recall that we introduced the method of time domain smoothing to specifically account for the unknown structure of $c(x)$. Hence, to have the term $c(x)$ cancel out, it must be that the asymptotic convergence speeds of the time and state domain estimators must coincide, again, up to a multiplicative constant. Starting with the state domain smoother, [Bandi and Moloche \(2018\)](#) argued that a *nearly* optimal bandwidth choice, given stationarity, is $h_{n,T} = C/\log(n)n^{1/d+4}$, which means that $nh_{n,T}^d = C^d n^{4/d+4}/\log(n)^d$. Additionally, the near optimal choice of bandwidth sequence $h_{n,T}$ removes the presence of the bias term in Theorem 2.2 asymptotically as argued by [Bandi and Moloche \(2018\)](#), meaning that the optimal sequence essentially clarifies the relevance of local adaptation in (non)stationary kernel regression in a way that the optimal bandwidth sequence should inversely be related to the number of available data points. And so if k_n denotes the number of data points used by the time domain smoother, and if $nh_{n,T}^d/k_n \rightarrow C > 0$ as

$n, k_n, T \rightarrow \infty$, then simple algebraic manipulation shows that

$$\frac{nh_{n,T}^d}{k_n} \sim \frac{n^{4/d+4}}{\log(n)^d k_n} \quad (2.4.1)$$

which consequently implies that $k_n = Cn^{4/d+4}/\log(n)^d$ for some constant $C > 0$.

Theorem 2.3. *Assume the conditions of Assumptions 1 to 3 and assume that X is stationary. If $nh_{n,T}^d/k_n \rightarrow C > 0$ and if K satisfies Assumption 4, then for any $t \geq 0$, the estimator $\hat{c}_{\text{TD},t}^n$ is independent to $\hat{c}_{\text{SD}}^n(x)$ as $n, k_n, T \rightarrow \infty$ conditional on $X_t = x$.*

Proof. Let us start by defining the truncated variant to $\hat{c}_{\text{SD}}^n(x)$. To do so, let ℓ_n be a sequence so that $\ell_n = o(n\Delta_n) = o(T)$ but $\ell_n \rightarrow \infty$ as $n, T \rightarrow \infty$. Then conditionally on $X_t = x$, we write

$$\hat{c}_{\text{SD},*}^n(x) = \frac{1}{\Delta_n} \frac{\sum_{j=1}^n \mathbb{1}_{|t-t_j| > \ell_n} K_{h_{n,T}}(X_{t_j} - x)(X_{t_j} - X_{t_{j-1}})(X_{t_j} - X_{t_{j-1}})^\top}{\sum_{j=1}^n K_{h_{n,T}}(X_{t_j} - x)},$$

in which the interpretation is that it is the ordinary Nadarya-Watson estimator, only it excludes any data X_s for which $|t - s| \leq \ell_n$ in the numerator. Now one readily verifies that for some $M > 0$ large, if $\ell_n > M$, then $\hat{c}_{\text{SD},*}^n(x)$ is estimated on data disjoint from that of $\hat{c}_{\text{TD},t}^n$ given that $k_n\Delta_n \rightarrow 0$ by assumption. In fact, the distance between the used observations of the two estimators is approximately ℓ_n . By Theorem A.1, X is ergodic as a consequence to stationarity and so by the results of Krengel (2011) along proposition 12.11 in Eisner et al. (2015), if X is stationary ergodic and if f is measurable, then $f(X)$ is stationary ergodic, which then means that $\hat{c}_{\text{SD},*}^n(x)$ and $\hat{c}_{\text{TD},t}^n$ are stationary ergodic. Hence if $\alpha(n)$ denotes the strong mixing coefficient, by Davydov's inequality of Davydov (1968) and the results of Yoshihara (1978), if p, q are so that $p^{-1} + q^{-1} < 1$ then

$$\begin{aligned} & \|\varphi_{\hat{c}_{\text{SD},*}^n(x) + \hat{c}_{\text{TD},t}^n}(\xi) - \varphi_{\hat{c}_{\text{SD},*}^n(x)}(\xi)\varphi_{\hat{c}_{\text{TD},t}^n}(\xi)\| \\ & \leq \alpha(\ell_n)^{1-p^{-1}-q^{-1}} C \|\exp(i \text{trace}(\xi^\top \hat{c}_{\text{SD},*}^n(x)))\|_p \\ & \quad \times \|\exp(i \text{trace}(\xi^\top \hat{c}_{\text{TD},t}^n))\|_q \end{aligned}$$

for every $\xi \in \mathbb{R}^{d \times d}$ where $\varphi_X(\xi) = \mathbb{E} \exp(i \text{trace}(\xi^\top X))$ denotes the characteristic function of a matrix valued random variable. Now the right hand side is bounded by $C\alpha(\ell_n)$ given that $|\exp(i \text{trace}(\xi^\top A))| \leq 1$ for every real compatible A . But it is known (see for example Orey (1971) or Meyn and Tweedie (2012)) that stationary ergodic processes which satisfy assumptions 1 to 3 are β -mixing, and so given that $\alpha(\ell_n) \leq \beta(\ell_n)$, it must be that $C\alpha(\ell_n) \rightarrow 0$ as $\ell_n \rightarrow \infty$. The measure-theoretic implications of the characteristic function imply the independence of $\hat{c}_{\text{SD},*}^n(x)$ to $\hat{c}_{\text{TD},t}^n$ conditional on $X_t = x$ as $n, k_n, T \rightarrow \infty$.

What is left to show is that $\mathbb{P}(\|\hat{c}_{\text{SD}}^n(x) - \hat{c}_{\text{SD},*}^n(x)\| > \varepsilon) \rightarrow 0$ for every $x \in \mathbb{R}^d$ and $\varepsilon > 0$ as $n, k_n, T \rightarrow \infty$. To do so, observe that

$$\hat{c}^* := \hat{c}_{\text{SD}}^n(x) - \hat{c}_{\text{SD},*}^n(x) = \frac{1}{\Delta_n} \frac{\sum_{j=1}^n \mathbb{1}_{|t-t_j| \leq \ell_n} K_{h_{n,T}}(X_{t_j} - x)(X_{t_j} - X_{t_{j-1}})(X_{t_j} - X_{t_{j-1}})^\top}{\sum_{j=1}^n K_{h_{n,T}}(X_{t_j} - x)}.$$

Now by continuity of sample paths and Assumption 1, there exists $C = C(\omega)$ such that $(X_{t_j} - X_{t_{j-1}})(X_{t_j} - X_{t_{j-1}})^\top \leq C|t_j - t_{j-1}|^2 = C\Delta_n^2$ for every $j = 1, \dots, n$. Moreover, K is a bounded function given that k is bounded continuous, and so for some $C' > 0$, it must be that $K \leq C'$ uniformly on $\mathbb{R}^d$. Lastly, theorem 3 in [Bandi and Moloche \(2018\)](#) ensures that if X is a stationary process, then $L_{n,T}(x)/T \rightarrow p(x)$ almost surely as $n, T \rightarrow \infty$ provided $\Delta_n/h_{n,T}^2 \rightarrow 0$ jointly with $n, T \rightarrow \infty$. Hence, by the Portmanteau lemma, $L_{n,T}(x)/T = p(x) + o_p(1)$ as $n, T \rightarrow \infty$, meaning that $L_{n,T}(x) = Tp(x) + o_p(T) \sim T$. Now $p(x) > 0$ for every $x \in \mathbb{R}^d$ by [Bogachev et al. \(2012\)](#) (see also Remarks 3 and 4) and so $\sum_{j=0}^n K_{h_{n,T}}(X_{t_j} - x) \sim p(x)T/\Delta_n = p(x)n$. Given that $\ell_n/\Delta_n = n\ell_n/T$ implies that $\sum_{j=0}^n \mathbb{1}_{|t-t_j| \leq \ell_n} = 2\ell_n/\Delta_n$, combining the above yields

$$\hat{c}^*(x) \leq \frac{n}{T} \frac{CC'\Delta_n^2 \sum_{j=0}^n \mathbb{1}_{|t-t_j| \leq \ell_n}}{\sum_{j=0}^n K_{h_{n,T}}(X_{t_j} - x)} = \frac{2nCC'\Delta_n^2 \ell_n}{p(x)nT\Delta_n} = \frac{2CC'\Delta_n \ell_n}{p(x)T}.$$

Now $2CC'\Delta_n \ell_n/p(x)T = 2CC'\ell_n/p(x)n$, yet ℓ_n was so that $\ell_n/n \leq \ell_n/n\Delta_n = o(1)$, which concludes the proof. $\square$

What is interesting here is that we have shown that two estimators, estimating the same quantity, are in fact independent provided that what they are estimating originates from a stationary process. But then, as in [Fan et al. \(2007\)](#), the assumption that $nh_{n,T}^d = Ck_n$ along with Theorem 2.3 and Corollary 2.1 must naturally imply that, conditional on $X_{t_j} = x$

$$\sqrt{nh_{n,T}^d} \text{vech}(\hat{c}_{\text{SD}}^n(x) - \hat{c}_{\text{TD},t_j}^n) \xrightarrow{d} \mathcal{N}(0, \Sigma(x))$$

provided that X is stationary, where $\Sigma(x) = (2\|k\|_2^{2d}/p(x) + C \text{diag} \text{vech}(V))P_D(c(x) \otimes c(x))P_D^\top$ is the sum of the variances of two estimators. Hence, if A is a $d(d+1)/2 \times 1$ vector of ones, then the mapping theorem tells us that

$$Z_j := A\Sigma^{-1/2}(x) \sqrt{\frac{2nh_{n,T}^d}{d(d+1)}} \text{vech}(\hat{c}_{\text{SD}}^n(x) - \hat{c}_{\text{TD},t_j}^n) \xrightarrow{d} \mathcal{N}(0, 1) \quad (2.4.2)$$

conditioned on $X_{t_j} = x$ for every $j \leq n$. Here, the existence of the inverse $\Sigma^{-1}(x)$ follows from the invertibility of $c(x)$ at every $x \in \mathbb{R}^d$ as a consequence to Assumption 2 and the strict positivity of $p(x)$ as a consequence to [Bogachev et al. \(2012\)](#). Yet, if the process is nonstationary, then

$$A\Sigma^{-1/2}(x) \sqrt{\frac{2nh_{n,T}^d}{d(d+1)}} \text{vech}(\hat{c}_{\text{SD}}^n(x) - \hat{c}_{\text{TD},t_j}^n) \rightarrow \pm\infty$$

almost surely, and so what we have created is an asymptotically valid method to assert whether a given trajectory is stationary at a single point t . What is difficult here is that, unlike ordinary hypothesis testing, we have n test statistics, namely one for each $t_j = \Delta_n, 2\Delta_n, \dots, T$. Although the exact asymptotic distribution of Z_j is known for each $j \leq n$, it is unknown what the joint distribution is of the vector $Z = (Z_j)_{j=1, \dots, n}$.

3 Constructing the Test

In this section, we will use the results above to actually construct a test statistic. To do so, we will use the sequence of test statistics obtained by means of Equation (2.4.2) where the goal is to find an expression for the probability that X is stationary on account of Z .

A first approach to test the global hypotheses of stationarity is to consider analysing n individual hypothesis, each having the form

$$H_0^{(j)} : Z_{t_j} \xrightarrow{d} \mathcal{N}(0, 1) \quad (3.0.1)$$

as $n, k_n, T \rightarrow \infty$. Consequently, the global hypothesis of stationarity can be assessed by means of multiple hypothesis testing frameworks, using the individual hypotheses of Equation (3.0.1). Given that Z_{t_j} is identically distributed for each j , perhaps the most rudimentary approach is the simple Bonferroni correction, which controls the type-I error at level α by solving $1 - (1 - \alpha_j)^n = \alpha$ for α_i the confidence level used in Equation (3.0.1). Yet the Bonferroni correction is often criticized on several grounds, including the restrictive condition of independence between hypotheses.

More sophisticated methods for multiple hypothesis testing are available, not dependent on any independence assumptions. An example includes the Benjamini-Hochberg procedure introduced by [Benjamini and Hochberg \(1995\)](#), which controls the false discovery rate (or FDR for short) under mild dependence assumptions. In essence, the procedure calculates the p -values for each obtained test statistic, orders them in an increasing order, say $p_{(j)}$, and rejects the global null hypothesis if

$$k_{\text{BH}} = \max\{i = 1, \dots, n : p_{(i)} \leq iq/n\}$$

exists with the convention $\max \emptyset = \infty$, where q is the FDR. [Benjamini and Hochberg \(1995\)](#) showed that their proposed test controls the FDR at level q , conditional on each hypothesis test being either independent or mildly related. Later on, [Benjamini and Yekutieli \(2001\)](#) introduced a more conservative version to the approach of [Benjamini and Hochberg \(1995\)](#), which is almost identical to the Benjamini-Hochberg procedure, only that we reject the global null if

$$k_{\text{BY}} = \max\{i = 1, \dots, n : p_{(i)} \leq iq/nH_n\} \quad (3.0.2)$$

exists, where $H_n = \sum_{j \leq n} j^{-1}$ denotes the n -th harmonic number. Contrary to the method of [Benjamini and Hochberg \(1995\)](#), the method of Equation (3.0.2) controls the FDR at arbitrary dependencies.

Remark 11. *It is worth remarking that, in general, the Benjamini-Hochberg and Benjamini-Yekutieli procedures simply give an upper bound for the false discovery rate, which is fundamentally different from the theoretical rejection rate that we wish to obtain for a test. However, in a simulated environment, if all individual null hypothesis are true, then the FDR coincides with the type-I error. This however generally does not work in applications.*

Some caution is in order when applying the above approaches. For example, it is well known that the Benjamini-Yekutieli approach is rather conservative in rejection of the global null hypothesis, and so controlling for the FDR comes with the trade-off in type-II errors. Indeed, if the precise dependence structure between each hypothesis test is known, then one could argue that the [Benjamini and Hochberg \(1995\)](#) methods is more robust, of course provided that the test agrees to the given set of conditions. Yet, generally, given the nonparametric nature of the problem, knowing the exact nature of the dependence structure of the obtained test statistics is hard. And so, assuming the null of stationarity, another take could be to use the properties of Gaussian processes. Given their substantial popularity in practice, many existing results are available with regards to the probabilistic aspects of Gaussian processes, which gives the opportunity to use existing literature to construct a new way of testing the stationary hypothesis.

3.1 The Running Maximum Test Statistic

The general idea here is to use a version of the Fisher-Tippett-Gnedenko theorem, a powerful result in the field of large deviation theory, which gives an asymptotic distribution of the running maximum of a sequence of random variables. Although this might seem disconnected from the discipline of hypothesis testing, the objective is as follows: suppose, under the null hypothesis, we should observe a sequence of variables $g_j \sim P_0$ where $j \in \mathbb{N}$ while the alternative hypothesis implies that $|g_j| \rightarrow \infty$ almost surely for every $j \geq 1$. Now if $\max_{j \leq n} |g_j|$ has a closed-form expression, then observing a process g_j which has a so-called running maximum exceeding the 95% critical value, we may argue that this is unlikely under the hypothesis that $g_j \sim P_0$, and so we reject the null. Yet, if the observed running maximum remains within the 95% critical region, then there is insufficient evidence to conclude that the null hypothesis is false.

To translate the Fisher-Tippett-Gnedenko theorem into the stationarity testing framework, recall that by the results of Section 2, stationarity of the process would imply that $Z_{t_j} \xrightarrow{d} \mathcal{N}(0, 1)$ for every $t_j = \Delta_n, \dots, T$ as $n, k_n, T \rightarrow \infty$. But then, if we can show that $Z = (Z_{t_j})_{j=0, \dots, n}$ converges to a centered Gaussian process with

unit variance, then we can use existing literature on the distribution of the running maximum to construct a single test statistic. This is done in two steps: first, we will show that indeed Z converges to a Gaussian process, after which we will extend on the Fisher-Tippett-Gnedenko theorem to be applicable to Gaussian process. For sake of brevity, the weak convergence is stated in Proposition A.3.

It is worth remarking that a general formula for the distribution of the running maximum of a Gaussian process requires some mild regulatory conditions. For example, Haan and Ferreira (2006) mentions that the Fisher-Tippett-Gnedenko theorem is a result applied to independent and identically distributed random variables, and so one could argue that, similar to the Billingsley CLT, some form of fading memory would be necessary to obtain a version applicable for mixing sequences. Even stronger, in the case of continuous time Gaussian processes, it is logical that Z_t must be almost perfectly correlated to Z_s for $|t-s|$ small to enforce some form of (stochastic) continuity. More precisely, Pickands (1969) argues that a necessity to assign a distribution to $\lim_{T \rightarrow \infty} \sup_{t \leq T} |Z_t|$ is that the covariance kernel $\gamma(s) = \text{Cov}(Z_t, Z_{t+s})$ of Z satisfies $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ as $t \rightarrow 0$ for $C > 0$ and $\alpha \in (0, 2]$ in addition to $\lim_{t \rightarrow \infty} \gamma(t) \log(t) = 0$. Here, the restriction $\alpha \leq 2$ is places to ensure that $\gamma(t)$ is positive definite as a consequence to the Bochner-Herglotz theorem (see, e.g., Thm. 18.6 in Eisner et al. (2015) or Remark 13 for a more exhaustive discussion). Yet, as we will see, simply retaining the original index of $X^n = (X_{k\Delta_n}^n)_{k=1, \dots, n}$ will inevitably imply the independence of Z_t with Z_s for any $s \neq t$ (c.f. Proposition A.4 and Corollary A.3), clearly violating this condition as $\alpha = 0$. Indeed, another approach could be to let Z^n be indexed by the natural numbers rather than the grid on which X^n is observed. However, Berman (1964, 1982) argues that in this case, the covariance kernel must satisfy $\gamma(k) = o(1/\log(k))$, for $k \in \mathbb{N}$ sufficiently large. Yet, by the construction of Z^n , projecting on $\mathbb{N}$ implies that there is guaranteed dependence within an open ball of radii $k_n/2 \rightarrow \infty$.

What is however possible is to artificially map the index Z^n onto $\mathbb{R}_0^+$. That is, suppose that X^n is observed at $\Delta_n, \dots, T$, then we construct $Z^n = (Z_{t'_j}^n)_{j=1, \dots, n}$ where $t'_j = t_j/\Delta_n k_n$, which will map $[t, t + k_n \Delta_n]$ onto $[t/\Delta_n k_n, t/k_n \Delta_n + 1]$, stretching the original index of Z^n by a factor $1/\Delta_n k_n \rightarrow \infty$. In essence, this will counter the vanishing covariance whilst keeping the convergence property to the centered Gaussian process, a proof can be found below Proposition A.3.

As a consequence, we can now focus on deriving the distribution of $\lim_{T \rightarrow \infty} \sup_{t \leq T} |Z_t|$. To do so, let us first introduce the formal hypothesis associated to the test, which is formulated as

$$H_0 : X \in \mathcal{P}_0(\mathbb{R}^d) \quad \text{versus} \quad H_1 : X \in \mathcal{P}_1(\mathbb{R}^d)$$

where $\mathcal{P}_0(\mathbb{R}^d)$ resp. $\mathcal{P}_1(\mathbb{R}^d)$ denote the sets of all stationary and nonstationary processes on $\mathbb{R}^d$ satisfying assumptions 1 to 3. Using the Gaussian process of Proposition A.3 along with the properties laid out in Theorem A.4, we can invoke the

theorems of [Pickands \(1969\)](#) as extensions to the works of [Berman \(1964\)](#), which give a closed form distribution to a normalised variant of $\lim_{T \rightarrow \infty} \sup_{t \leq T} |Z_t|$ on account of the aforementioned restrictions. For completeness, the result is stated in [Theorem A.3](#).

Here, as a consequence to [Haan and Ferreira \(2006\)](#), the limiting distribution is the standard Gumbel distribution $F(x) = \exp(-e^{-x})$, which is strictly increasing with inverse $F^{-1}(x) = -\log \log x^{-1}$ on $(0, 1)$. Consequently, using the distribution of [Theorem A.3](#), if $\phi_n = \max_{k \leq n} |Z_{t'_k}|$ denotes the global test statistic, then we reject the null hypothesis at level $\alpha \in (0, 1)$ if

$$\phi_n > b_n - \frac{\log \log \alpha^{-1}}{a_n} \quad (3.1.1)$$

where $a_n = \sqrt{\log(n/k_n)}$ and

$$b_n = a_n + \frac{(2 - \xi)/\xi \log \log(n/k_n) + 2 \log((2\pi)^{-1/2} C^{1/\xi} H_\xi 2^{(2-\xi)/2\xi})}{2a_n}.$$

Here, C and ξ denote the coefficients of [Theorem A.3](#). In the special case that $\xi = 1$, [Pickands \(1969\)](#) showed that $H_1 = 1$ while $\xi = 2$ corresponds to $H_2 = 1/\sqrt{\pi}$.

3.1.1 Adaptation To Finite Samples

Although the theory aforementioned essentially implies the validity of the test in an asymptotic framework, in practice, it is desirable to apply the method only when a finite sample is available. Yet, [Berman \(1964, 1982\)](#) and [Giné and Nickl \(2021\)](#) argue that the order of convergence of the running maximum is $1/\log(T)$, meaning that a considerable amount of data is required to obtain reliable estimates using the results of [Pickands \(1969\)](#). As highlighted by [Hall \(1979\)](#), the order of convergence can be improved to $T^{-\delta}$ by adjusting the asymptotic bounds for some $\delta > 0$, although this is generally hard and requires thorough knowledge on the exact structure of Z , see for example the comments in [Giné and Nickl \(2021\)](#). Another strain is that the results of [Berman \(1964, 1982\)](#) are obtained for stationary centered Gaussian processes in continuous time. Although we asymptotically converge (weakly) to a continuous time process, in finite samples, the obtained sequence of test statistics is discrete and merely càdlàg rather than continuous.

A practical solution is to consider the distribution associated to the running maximum of a discrete Gaussian process. For example, [Berman \(1964\)](#) showed that a sequence of independent centered Gaussian random variables $\xi = (\xi_n)_{n \geq 1}$ with unit variance satisfies

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(a_n (\sup_{k \leq n} |\xi_k| - b_n) \leq x \right) = \exp(-e^{-x}) \quad \forall x \in \mathbb{R}$$

where $a_n = \sqrt{2 \log(n)}$ and $b_n = a_n - (\log \log(n) + \log(\pi))/2a_n$. However, the obtained Gaussian sequence Z^n does not constitute an i.i.d. sequence as there is

significant overlap in the data used to construct the test statistic, especially in adjacent test statistics which are virtually fabricated on equal data making them severely dependent. This makes the application of the confidence bounds obtained by [Berman \(1964, 1982\)](#) unlawful. Fortunately, it can be shown that Gaussian processes with vanishing correlations still agree to a closed form distribution of the asymptotic running maximum. What is more is that this limit coincides with the case of an i.i.d. sample of standard normal random variables and does not require any conditions on the autocovariance function $\gamma(n)$ for small n . In particular, if ξ denotes a stationary sequence of standard normal variables satisfying $\lim_{n \rightarrow \infty} \log(n)\gamma(n) = 0$, then the result of [Berman \(1964, 1982\)](#) remains valid, a proof is given below Proposition [A.2](#). Indeed, as aforementioned, the condition that $\gamma(n) = o(1/\log(n))$ is not satisfied asymptotically, although it would still be interesting to assert the behaviour of the test in finite samples, see also Section [3.2](#) for a more exhaustive discussion. As a consequence, in finite samples, the null hypothesis of stationarity should be rejected if $\phi_n = \max_{k \leq n} |Z_{t_k}^n|$ satisfies Equation [\(3.1.1\)](#), where $a_n = \sqrt{2 \log(n)}$ and $b_n = a_n - (\log \log(n) + \log(\pi))/2a_n$, replacing the coefficients of [Pickands \(1969\)](#).

As a last remark, although existing literature on large deviation principles for Gaussian processes is rich in advanced and better suited approximations to $\sup_{t \leq T} |\xi_t|$, these are generally technical and so we leave any improvements for future research. For inspiration, we recommend the theory and references in [Friz and Victoir \(2010\)](#), [Haan and Ferreira \(2006\)](#), [Kallenberg \(1997\)](#) and [Giné and Nickl \(2021\)](#).

3.2 Remarks and Shortcomings

In this section, we will discuss some practical issues and shortcomings of the proposed method. In particular, we will discuss various difficulties regarding the bandwidth sequences, the ratio of used data between the state and time domain estimators, and the convergence speeds of the running maximum test statistic.

To start, the choice for an optimal bandwidth sequence, although known up to a multiplicative constant, can cause for invalid inferential results. That is, although the proposed near optimal bandwidth sequence of [Bandi and Moloche \(2018\)](#) ensures smooth convergence of the estimator as $n, T \rightarrow \infty$, an uninformed choice of the multiplicative constant can render the test invalid in small samples. This becomes especially evident if we look at the convergence speed of the optimal bandwidth $h_{n,T} = C/n^{1/d+4} \log(n)$. Namely, by the slow convergence rate of $h_{n,T}$, large amounts of data are needed to counter an adverse choice of C . Moreover, besides the choice to the constant C , the choice to exclude the bias $\Gamma(\cdot)$ in the equation to construct the test statistic can lead to invalid results. That is, as argued by [Bandi and Moloche \(2018\)](#), the optimal choice for $h_{n,T}$ merely vanishes the bias as $n, T \rightarrow \infty$, yet the influence of $\Gamma(\cdot)$ can be severe in small samples, where an uninformed choice to C

can exacerbate this phenomenon.

Another practical issue is that, if the process were to be nonstationary, then the invariant probability measure does not exist, yet the asymptotic variance of the test statistic is constructed using the normalised invariant measure, which in the nonstationary case would be zero everywhere. That is to say, the origin of the invariant density was due to $L_{n,T}(x)/T \rightarrow p(x)$ in probability, yet this fails if X is nonstationary, as [Bandi and Corradi \(2014\)](#) pointed out that $L_{n,T}(x)/T \rightarrow 0$ almost surely as $n, T \rightarrow \infty$ in this case. Fortunately, by the theorem of [Darling and Kac \(1957\)](#), an alternative scaling can be obtained by means of the occupation measure $L_{n,T}$. Namely, [Bandi and Moloche \(2018\)](#) showed that for any Harris recurrent process X satisfying assumptions 1 to 3 with regularity index α

$$\frac{L_{n,T}(x)}{U(T)T^\alpha} \rightarrow C_X \tilde{\phi}(x) g_\alpha$$

in the weak topology as a consequence to the Darling-Kac theorem of [Darling and Kac \(1957\)](#), where U is again a Karamata slowly varying function, C_X a process specific constant, g_α denotes the Mittag-Leffler density with parameter α and $\tilde{\phi}(x) = \phi(dx)/dx$. Hence, rather than to standardise using the invariant density $p(x)$, we can standardise using $C_X \frac{\phi(dx)}{dx} g_\alpha$, which can be computed using standard nonparametric techniques. As noted in [Bandi and Moloche \(2018\)](#), this should in practice be done in an automated way, where one first infers the order of $U(T)T^\alpha$ after which the constant can be obtained. To give an example, if X were to be the standard Brownian motion, then the Kallianpur-Robbins theorem states that $\alpha = 1/2$ and $L_{n,T}(x) = O_p(\sqrt{T})$ with $U(T) = 1$. Consequently, we would first need to infer the approximate size of α , after which we can estimate the multiplicative constant associated to the divergence of $L_{n,T}(x)$. This would then give reliable results in a simulated environment, although in practice this could be rather tricky, especially given that U needs only to be Karamata slowly varying at infinity. A way to mitigate this is by first normalising the stochastic process, which in theory should reduce the range in which C_X takes values. Yet, [Bandi and Moloche \(2018\)](#) and [Darling and Kac \(1957\)](#) argue that, especially in a finite sample, the behaviour of C_X is dependent not only on the regularity index, but additionally on the choice of kernel K and the bandwidth sequence $h_{n,T}$.

Besides the configuration, the method of stationary testing based on the running maximum additionally comes with various shortcomings. To start, As for the finite sample adaptation of the proposed test statistic, although this will considerably mitigate the complexity of the confidence bounds of [Pickands \(1969\)](#), it is worth remarking that the obtained discrete process indexed by the natural numbers does not satisfy the condition of $\gamma(k) = o(1/\log(k))$ as $k \rightarrow \infty$. That is to say, for sufficiently small Δ_n and large n, T , the choice for the continuous confidence bounds should be taken rather than the discrete adaptation. Yet, as argued by [Giné and Nickl \(2021\)](#) and [Berman \(1982\)](#), the point at which this change should be enforced inevitably

depends on the exact nature of dependence, which as of now is unknown. Another complication is that the results of [Berman \(1964, 1982\)](#) and [Pickands \(1969\)](#) are in an asymptotic framework, and so an empirical test essentially argues its validity using arguments that hinge on large sample statistics, where a large sample size usually guarantees a small bias. However, as opposed to regular testing, the running maximum test statistic is based on the extremes of a sequence. As a consequence, a probable issue is that the tail estimates can be far from the actual tail behaviour of the asymptotic result, even if the difference between the two measures is small (for example measures with respect to the Hellinger distance or the Kullback-Leibler divergence). Said another way, the uncertainty in inferring the tail behaviour of an extreme event can be severe, even if the sample size is reasonably large. Indeed, in case of ordinary statistical inference, in small samples, a general practice is to construct the critical region by means of the student- t distribution rather than the normal distribution to account for the additional uncertainty. However, [Haan and Ferreira \(2006\)](#) argue that, in case of extreme value theory, the Pickands-Balkema-De Haan theorem (along with the generalised Fisher-Tippett-Gnedenko theorem) provides a finite list of possible limiting distributions, which to the best of our knowledge has no t -distribution counterpart in case of small samples.

A last remark comes from a numerical point of view. Namely, as we are comparing the standardised difference of the (half) vectorised estimators, a simple numerical flaw, for example obtained from inaccurate estimates when inverting the asymptotic variance, could cause for a spike in the sequence of test statistics. This inevitably will result in rejection of the null hypothesis, even if the underlying process is stationary. Generally, the likelihood of these events is severely dependent on the aspects of the underlying process, and so any potential future research could try to find methods that come with these types of inconsistencies.

It is worth remarking that, although this framework is centered around continuous time diffusion processes of Itô's type, it is fairly easy to construct the same procedure for discrete processes, in particular discrete Markov processes of the recurrent type. That is to say, as is for example shown in [Meyn and Tweedie \(1993\)](#) and [Geman and Horowitz \(1980\)](#), the results of the Darling-Kac theorem do apply to their discrete counterparts in the sense that for a particular set of functions f , the growth rate of the partial sum $S_n = \sum_{k=1}^n f(X_k)$ of a (Harris) recurrent Markov chain depends on its regularity index. Hence, constructing a test statistic can virtually be done using the same toolbox with some minor adjustments to account for the discrete nature of Markov chains. However, both in the discrete as the continuous time, a necessary condition must always be that the state domain is unbounded given that the existence of an invariant probability measure is guaranteed by the Krylov-Bogoljubov theorem (c.f. §I, Ch. 5, B.3 in [Granas and Dugundji \(2003\)](#) or Thm. 5.1 in [Eisner et al. \(2015\)](#)) if the state space were to be compact.

4 Simulation Study

In order to infer the finite sample properties of the constructed test, we will simulate three bivariate diffusion processes using the Euler-Maruyama scheme with the Milstein correction, which improves the approximation of the stochastic differential equation (Bishwal (2007)). In doing so, we wish to assert how the proposed method applies when practiced on a simulated model, which is known to be stationary or not beforehand, where the main interest is in the observed rejection rates in comparison to the theoretical expected rejection rates, but also the properties of convergence when additional data is added.

4.1 Models and Configuration

To start, we use the ubiquitous stationary Ornstein-Uhlenbeck process (or Mehler semigroup), which, by arguments in i.a. Bandi and Corradi (2014); van der Vaart and Wellner (2013), is a Gaussian convolution with time-scaling and can be considered to be the continuous-time counterpart of the invertible autoregressive process $x_t = \rho x_{t-1} + \varepsilon_t$ where $(\varepsilon_t)_{t \geq 1}$ is an independent and identically distributed sequence of standard normal random variables and $|\rho| < 1$. We use the notation of Revuz and Yor (1999) to define the unique strong solution

$$X_t = e^{-\kappa t} X_0 + A e^{-\kappa t} \int_0^t e^{\kappa s} dW_s \quad \forall t \geq 0$$

for $\kappa > 0$ and $A \in \mathbb{R}$ with $X_0 \sim \nu_0$ the invariant measure. Indeed, as in Bandi and Corradi (2014), this model is stationary (ergodic) if and only if $\kappa > 0$, whilst the case of $\kappa = 0$ corresponds to a scaled Brownian motion. Therefore, under the assumption of correct specification, in practice, one could use parameter inference on κ (see Ait-Sahalia (1996) and Bishwal (2007)) to infer the stationary properties of an Ornstein-Uhlenbeck process, yet it would be interesting to complement this endeavour with a stationarity test, to either confirm the results of the specification test, or to conclude that the methods yield opposing conclusions.

Next, we consider a planar Brownian motion $W = (W_t)_{t \geq 0}$ which, conditional on $W_0 = 0$ $\mathbb{P}$ -a.s., follows a Gaussian bivariate law with mean zero and variance tI_2 converging to the two-dimensional Lebesgue measure $m_2 : (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2)) \rightarrow [0, \infty]$ as $t \rightarrow \infty$. As aforementioned, given that the invariant measure is merely σ -finite, we can not standardise the obtained difference of the state and time domain estimators using the invariant density. To overcome this, we can analyse the process specific constant of the Darling-Kac theorem, which is simply a proportionality constant to obtain the desired Mittag-Leffler distribution. A quick simulation study showed that the simulated trajectories, upon normalisation with $\log T$, are of order 50, meaning that $C_X = 1/50$. That given, we can hypothesise that the state domain estimator should only have the central limit property with rate proportional to

$\sqrt{h_n^2 \log(T)/\Delta_n}$, meaning that by enforcing the linear growth rate, the limiting distribution should have a variance proportional to $T/\log(T)$ as a consequence to Equation (2.4.1).

It is worth remarking that, by arguments in [Revuz and Yor \(1999\)](#) or [Gettoor \(1980\)](#), these convergence rates are only asymptotically valid, meaning that in smaller samples, it can be hard to distinguish the stationarity properties on account of the convergence rates to achieve the central limit property, as was the case in ([Bandi and Corradi, 2014](#), Tab. 3). What is more is that on small samples, a planar Brownian motion can exhibit stationary and ergodic type of behaviour, even though the occupation measure is asymptotically dominated by any positive function growing faster than $\log(T)$, which could render the test invalid if this occurs frequently enough. Yet, we would expect an increase in the observed rejection rate as T grows.

Lastly, we consider a time-inhomogeneous Itô semimartingale $X = X_0 + A + M$ with local martingale σW for W a correlated Brownian motion ($\rho = 1/2$) and $\sigma = (\sigma_t)_{t \geq 0}$ a progressive process defined by $\sigma_t = \alpha_1 + \alpha_2 \sin(16\alpha_3 \pi t/n)$ with $(\alpha_1, \alpha_2, \alpha_3) = (2.1, 0.5, 1)$, and $A(x, y)dx = -0.1ydx$. Here, if $\alpha_3 = 0$ or $\alpha_2 = 0$, then X would in fact be characterised by a stationary ergodic Ornstein-Uhlenbeck process, and so it is interesting to see whether a slight time-varying trend in the volatility would result in rejection of the null hypothesis. From its construction, by definition of the test, we can hypothesise that the time-inhomogeneous nonlinear Markov process will yield a rejection rate close to one as a consequence to the violation of the ergodic hypothesis in the Birkhoff-Khinchin theorem. As a small note, given that the process is time-induced nonstationary, locally, we may still expect the process to be distributed like an Ornstein-Uhlenbeck process. What is more is that the averaged kernel $L_{n,T}(x)$ will likely grow at a linear rate for each $x \in \mathbb{R}^2$, hence, we include the invariant measure of an Ornstein-Uhlenbeck process in the weak limit to solely infer the consequences of time-varying volatility.

Given that the asymptotic distribution of the test statistic is known by virtue of the results in [Berman \(1964, 1982\)](#) with the adaptation of Proposition A.2, we will investigate the simulated rejection rates if $\alpha \in \{1\%, 5\%, 10\%\}$, where we let $n \in \{1000, 3000, 7300\}$, $\Delta_n = 1/20$, meaning that $T \in \{50, 150, 365\}$, i.e. 50, 150 and 365 days of stock data on 18 minute intervals based on 6 hour trading days, which by [Bandi and Corradi \(2014\)](#) are common in the financial literature. As for the parameter configuration of the test statistic, following some minor numerical improvements, we let $h_{n,T} = C_n/n^{1/6} \log(n)$ where $C_n = 2$ if $T = 50$, $C_n = 6$ if $T = 150$ and $C_n = 9$ if $T = 365$. For k_n , we take $k_n = C'_n n^{2/3} / \log(n)^2$ for $C'_n = 20$ for every n . As for the kernel K , we choose $k_h(x) = k(x/h) = \mathbb{1}\{|x|/h \leq 1\}$ with $K_h(x) = \frac{1}{h^d} \prod_{i=1}^d k_h(x^i)$, which is in $L^p(\mathbb{R})$ for every $p \geq 1$ and satisfies the regularity conditions of [Bandi and Moloche \(2018\)](#) except for the continuity property, although for finite samples this should not pose any problems. In fact, this is the kernel used by [Brugière \(1993\)](#) and in literature is referred to as the indicator kernel or

generalised boxcar kernel, which has many welcoming properties and is often simple to implement and understand. However, its simplicity also comes with various limitations and so it would be interesting to assert the small sample properties under various alternatives such as the Epanechnikov kernel or the Silverman kernel.

4.2 Results

To evaluate the results using the running maximum test statistic, we will compare the outcomes with those obtained using the Benjamini-Hochberg and Benjamini-Yekutieli procedures on account of their empirical size and power. Given that existing literature on nonparametric stationarity testing assumes that the process is nonstationary under the null hypothesis, i.e., in [Bandi and Corradi \(2014\)](#) and [Kanaya \(2011\)](#), it is generally not possible to compare the characteristics of our method with respect to their approach. Therefore, we will compare our method to the ubiquitous Kwiatkowski-Phillips-Schmidt-Shin test (KPSS) and the lesser known Leybourne-McCabe (LM) test. In essence, both methods are constructed in the realm of linear discrete time series models of the autoregressive class. Therefore, we can expect that both methods will correctly reject the null of stationarity in case of a planar Brownian motion, but will have trouble characterising the stationarity properties of nonlinear Markov processes. In¹ [Table 1](#), the rejection rates are shown at a 5% confidence level, in which standard deviations are calculated according to the Bernoulli sample estimator variance $\hat{\sigma}_n^2 = \hat{p}(1 - \hat{p})/n$ for $\hat{p}$ the (estimated) rejection rate and $n = 200$ the number of independent draws. Indeed, the rejection rate sample variance is biased given that it does not include the degrees of freedom correction, yet for $n = 200$, this difference is negligible. In [Appendix A.4](#), additional results are provided for the 1% confidence level in [Table 2](#) and the 10% confidence level in [Table 3](#).

From [Table 1](#), starting with the simulated Ornstein-Uhlenbeck processes, we find that for $T = 50$ days, the running maximum rejected 10% of the simulated paths on grounds of their stationarity properties. The same rejection rate is achieved by the Benjamini-Hochberg procedure while the Benjamini-Yekutieli approach was more conservative in rejection of the stationary hypothesis. Interestingly, we see that both the KPSS and especially the LM tests conclude that the majority of simulated sample paths are nonstationary. Given that the Δ -skeleton (obtained by the Euler-Maryuama discretization) of an Ornstein-Uhlenbeck process is an autoregressive process of order one, we would expect that the KPSS and LM tests should obtain rejection rates close to the (asymptotic) theoretical rate of 5%. In fact, the sampled chain $X^{\Delta_n} = (X_{k\Delta_n})_{k \in \mathbb{N}}$ would correspond to an autoregressive process with coefficient $\rho = e^{-\kappa\Delta_n}$ (c.f. exercise 20.21 in [Friz and Victoir \(2010\)](#) and example 10.4 (a) in [Bishwal \(2007\)](#)). Plugging in the chosen values $\kappa = 1/5$ and $\Delta_n = 1/20$ shows that

¹Following the comments in [Section 3](#), we employed the discrete version to the [Berman \(1982\)](#) confidence bounds rather than the continuous version.

Bivariate Ornstein-Uhlenbeck					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	5%	5%	5%	5%	5%
50	0.100 (0.0004)	0.100 (0.0004)	0.075 (0.0003)	0.690 (0.0011)	0.965 (0.0002)
150	0.060 (0.0003)	0.060 (0.0003)	0.015 (0.0001)	0.565 (0.0012)	0.860 (0.0006)
365	0.075 (0.0003)	0.080 (0.0004)	0.055 (0.0003)	0.355 (0.0011)	0.740 (0.001)
Planar Brownian motion					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	5%	5%	5%	5%	5%
50	0.280 (0.001)	0.300 (0.001)	0.175 (0.0007)	0.970 (0.0001)	0.990 (0.000)
150	0.695 (0.0011)	0.720 (0.001)	0.615 (0.0012)	0.975 (0.0001)	0.990 (0.000)
365	0.955 (0.0002)	0.980 (0.0001)	0.895 (0.0005)	1.000 (0.000)	1.000 (0.000)
Bivariate inhomogeneous diffusion					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	5%	5%	5%	5%	5%
50	0.315 (0.0011)	0.335 (0.0011)	0.190 (0.0008)	0.620 (0.0012)	0.925 (0.0003)
150	0.225 (0.0009)	0.235 (0.0009)	0.120 (0.0005)	0.545 (0.0012)	0.845 (0.0007)
365	0.165 (0.0007)	0.180 (0.0007)	0.125 (0.0005)	0.340 (0.0011)	0.675 (0.0011)

Table 1: Rejection rates of the Euler-Maruyama simulated diffusion processes with the Milstein correction for $T \in \{50, 150, 365\}$, $n \in \{1000, 3000, 7300\}$ and $\Delta_n = 1/20$ using a version of the Fisher-Tippett-Gnedenko running maximum theorem, the Benjamini-Hochberg and Benjamini-Yekutieli procedures and the KPSS and LM tests.

$\rho \approx 0.99$, which corresponds to a so-called near integrated process. Given that the KPSS and LM tests essentially work by asserting if $\rho = 1$, the empirically obtained ρ would likely require a large sample size to be able to significantly distinguish it from one, which partially explains the high empirical size.

Increasing the sample size to 150 days shows that the proposed method converges to the theoretical 5% rejection rate, for which the running maximum and Benjamini-

Hochberg procedure achieve an empirical rejection rate of 6%. Again, as above, the Benjamini-Yekutieli procedure is much more reluctant to reject the global null hypothesis of stationarity. A final increase to 365 days shows that the empirical rejection rates increased slightly. An explanation for this phenomenon could be that increasing T whilst letting Δ_n be fixed can cause the test to obtain invalid results, given that Δ_n should converge to zero as T goes to infinity. We additionally find that the observed rejection rates of the KPSS and LM tests decrease, further strengthening the claim above.

As for the simulated paths of a planar Brownian motion, we see that the running maximum approach is quick to recognise the nonstationary characteristic of a Brownian motion. Indeed, as discussed in [Bandi and Corradi \(2014\)](#), increasing the number of trading days corresponds with a rapid convergence to an empirical rejection rate of 100%. This aligns with the theory of Section 3, given that the test abuses the discrepancy in divergence rates, which will become more evident as $n, T \rightarrow \infty$.

Comparing the rejection results with the Benjamini-Hochberg and Benjamini-Yekutieli approaches shows that the running maximum construction works in practice, even for small samples. However, in case of a planar Brownian motion, we see that our tests, for all choices of T is unable to beat the empirically obtained rejection rates from the KPSS and LM tests, both tailored for identifying or rejecting the stationary hypothesis in linear models. In fact, as a consequence to Eq. (14.5.6) in [Davidson \(2000\)](#), the test statistic diverges with rate n under the alternative, which clearly aligns with our findings and basically guarantees a 100% rejection rate for unit-root Δ -skeletons in sufficiently large samples.

At last, for a bivariate inhomogeneous diffusion process, we see that increasing T decreases the rejection rate for the null of stationarity. Indeed, here, it is not the divergence rate of the occupation measure which is the identifying characteristic of nonstationarity, but rather the discrepancy in the ergodic hypothesis caused by the bias in the state domain smoother. That is to say, by [Eisner et al. \(2015\)](#), the results of Appendix A and [Lazić and Sandrić \(2021\)](#), a stationary process (hence ergodic under assumptions 1 to 3) must obey the “state mean equals time mean” property. More precisely, for $\mathbb{P}$ -a.e. $\omega' \in \Omega_0$, the sample space, and $u > 0$

$$\lim_{T \rightarrow \infty} \int_0^T X_t(\omega') dt = \int_{\Omega_0} X_u(\omega) d\mathbb{P}(\omega),$$

yet if X is time inhomogeneous, then this condition fails for some $u > 0$. Indeed, the results of Table 1 show that the proposed test can to an extent filter the nonstationary nature of time inhomogeneous processes, yet the results do clearly indicate that an increase in T whilst leaving Δ_n fixed reduces the power. Given the construction of the process, this can be explained by noticing that, as T grows, locally, the process starts to resemble a stationary ergodic Ornstein-Uhlenbeck process. Hence, rather

than finding a discrepancy in the speed of convergence, here, the key is to include as much data as possibly available to find discrepancies in the estimated diffusion coefficients. It is therefore reasonable that the effective sample size is dependent on how the time varying structure relates to the other characteristics of the process. For example, a rapidly oscillating component whose amplitude is significantly larger than the local volatility is likely to require much less data to detect nonstationarity as opposed to processes that have underlying time-dependent components which move slow compared to the drift and diffusion coefficients.

5 Financial Data

Financial data, especially in the realm of high-frequency, is often assumed to come from a diffusion process. For example, [Ait-Sahalia \(1996\)](#) argues that if X were to be a limited liability asset, then popular choices for models would include a Cox-Ingersoll-Ross model, a positive Ornstein-Uhlenbeck process or a Brownian motion with drift. Depending on the context, the choice of what model to deploy can have significant impact on the validity of the results, which can be sensitive in financial situations. For example, a market maker might use a simple Bachelier stochastic differential equation or a Black-Scholes Model given that on a small scale, no real drift is present and volatility can in general be taken fixed on small intervals, see for example [Guéant \(2016\)](#). However, large investment firms, seeking to generate revenue by means of (long-term) mean-reversion strategies, are arguably more reluctant to selecting a simple Bachelier or Black-Scholes model over more adapted and customised alternatives.

To stick with the example of mean-reversion, say that we are interested in investigating whether a portfolio, composed of two assets, is suitable for mean-reversion strategies. Crudely speaking, a mean-reverting asset is characterised by, as the name suggests, mean-reverting behaviour, which usually intensifies as the asset moves away from its theoretical mean. An example may include a pair of assets, who individually might be nonstationary, but their difference follows a stationary process (in discrete linear models also called cointegration), although a portfolio with two independent Ornstein-Uhlenbeck processes still constitutes a mean-reverting portfolio. [Guéant \(2016\)](#), among others, state that, in case of mean-reversion, an Ornstein-Uhlenbeck model is a good first choice to exercise on the given portfolio, although more appropriate models such as the Kessler-Larsen-Sørensen Models or a Jacobi Diffusion Model are additionally known to be applicable in financial applications (for a discussion, see, e.g., [Bishwal \(2007\)](#)). To simulate this scenario, we consider the two assets Polygon against Stellar (MATIC/XLM) and SushiSwap against PanCakeSwap (SUSHI/CAKE), both known examples of mean-reverting pairs. We compare the results with the, likely to be nonstationary, portfolio composed of Render Token and Fetch.AI (abbreviated to RNDR and FET), both expressed in Tether

Token (USDT).² The data was obtained using the CryptoCompare API and is composed of historical (mid-price) data sampled at the hourly frequency from January 1st 2024 up until and including June 16th 2024. Given that the crypto exchange is a 24/7 financial market, this amounts to $\Delta_n = 1/24$ with $T = 166$ and $n = 3986$.

In Figure 1, the two portfolios are shown. Initially, the left graph seems to resemble the behaviour of a natural scale diffusion process $dX_t = \sigma(1 + X_t^2)^\gamma dW_t$, where $\gamma > 1/2$ corresponds to a so-called volatility induced stationary process having an invariant density $p(dx) = dx/(1 + x^2)^{2\gamma}$. Indeed, we see that, although considerable spikes are present, these are then quickly reverted back to the mean. Yet, focusing on the right panel showing the RNDR and FET processes, we see that the sample paths much resemble a correlated Brownian motion process, arguably with time dependent diffusion functions, which is well known to be nonstationary.

5.1 Configuration

Before we start, we need to set the configuration for the test. That is, we need to decide on the values of the bandwidth sequence and the ratio of observations used in the time versus the state domain estimators.

We start with the choice of the bandwidth. Given that our method relies on the results of [Bandi and Moloche \(2018\)](#), we will use their proposed *nearly* optimal bandwidth choice $h_{n,T} = C/n^{1/6} \log(n)$, which eliminates the bias of the estimator and is compatible with the time domain smoother following Equation (2.4.1). However, and especially in small samples, the choice for C can cause the test to be either too conservative or too radical in its decision on the plausibility of stationarity. What is more is that the optimal choice of C (with respect to some prior chosen metric) depends on the underlying process. For one, a process that has state dependent volatility arguably requires a much smaller value of C than a process with constant volatility. In fact, [Bandi and Moloche \(2018\)](#) demonstrate that if $c(x) = c \in \mathbb{R}$ (or $\mathbb{R}^{2 \times 2}$) irrespective of x or $b(x)$, then a non-vanishing bandwidth would obtain asymptotically valid results, given that $\hat{c}$ converges to c by the ergodic theorem along with the mapping theorem.

To infer the optimal choice for C , we run a simulation to assess the performance of the estimator $\hat{c}_{SD}^n(x)$ using the optimal bandwidth of [Bandi and Moloche \(2018\)](#),

²Here, the choice of Tether as opposed to other crypto assets is to investigate the (non)stationarity of Render Token and Fetch.AI, given that Tether is a so-called stable asset, closely following the US-dollar. Given that the value of the dollar is in general a reasonable estimate for inflation, investing in Tether is considered to mirror the act of not investing, showing that USDT is strongly mean reverting around one. Therefore, expressing the value of a crypto in terms of USDT essentially reflects the true value of the asset, with a slight perturbation caused by small fluctuations in the difference between Tether and the dollar.

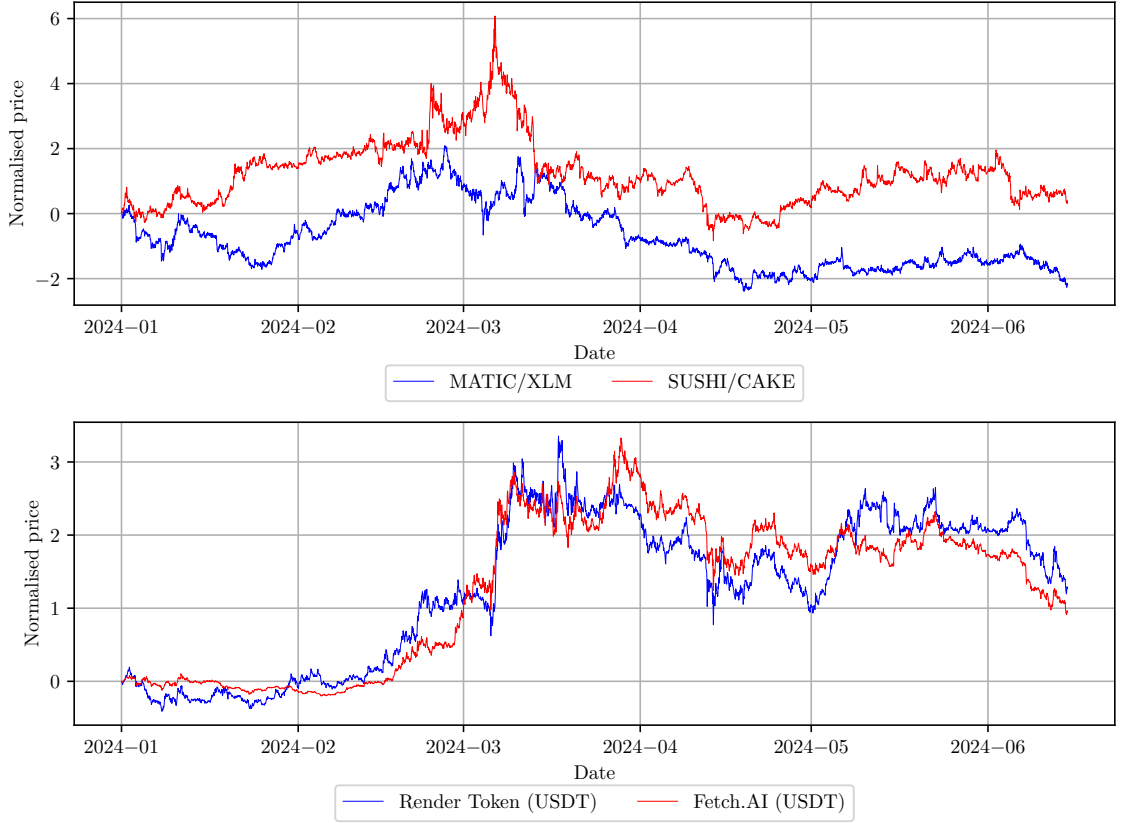


Figure 1: Sample paths of the empirically observed crypto assets. The upper panel depicts the normalised paths of MATIC against XLM (blue) and SUSHI against CAKE (red). The lower panel shows the normalised paths of Render Token (blue) and Fetch.AI (red), both expressed in terms of Tether.

measured with respect to the average prediction error

$$\text{APE}(\hat{c}_{\text{SD}}^n) = \frac{1}{n} \sum_{j=1}^n \text{trace}(Y_{t_j} Y_{t_j}^\top - \hat{c}_{\text{SD}}^n(X_{t_{j-1}}))^2$$

where $Y_{t_j} = (X_{t_j} - X_{t_{j-1}})/\sqrt{\Delta_n}$. At first, given that the average prediction error is attuned to optimise models primarily used for forecasting purposes, it seems unintuitive to select a bandwidth sequence based on the model's ability to forecast. However, [Fan et al. \(2007\)](#) showed that, although more appropriate methods are available, such as the entropy loss $\ell_1(\hat{c}) = \sum_{j=1}^n \text{trace}(c_{t_j}^{-1} \hat{c}_{t_j}) - \log \det(c_{t_j}^{-1} \hat{c}_{t_j}) - \dim(\hat{c}_{t_j})$ or the quadratic loss $\ell_2(\hat{c}) = n^{-1} \sum_{j=1}^n \text{trace}(c_{t_j} - \hat{c}_{t_j})^2$, these methods inevitably rely on the unknown shape of c_t . Moreover, the prediction error tends to regulate the estimated volatility, reducing the likelihood to overfit. In fact, in a parametric setting, minimising the least squares objective function is often a reliable approach in obtaining parameter estimates, see for example [Bishwal \(2007\)](#).

Given that the empirical data has a length of $T = 166$ with $n = 3986$ and $\Delta_n = 1/24$, letting $C = 1$ yields $h_{n,T} = 0.02$. If $C \geq 25$, then over 50% of data would be used

to calculate the diffusion coefficient for the state domain method. Hence, we will check the average prediction error with $C \in \{i = 1, \dots, 25\}$. Doing so showed that the model exercised on the first portfolio, i.e., the set containing MATIC/XLM and SUSHI/CAKE, achieved an optimal bandwidth $h_{n,T} = 9/n^{1/6} \log(n) \approx 0.27$, obtaining an average prediction error of 0.8. As for the second portfolio, i.e., the set containing Render Token and Fetch.AI, the optimal bandwidth was again $h_{n,T} = 9/n^{1/6} \log(n) \approx 0.27$, this time achieving an average prediction error of 0.05.

Lastly, we need to decide on the ratio between the number of data points used by the state domain estimator and the time domain smoother. Indeed, as aforementioned, the ratio should asymptotically be $k_n = Cn^{2/3}/\log(n)^2$ where n denotes the number of total observations and k_n denotes the number of points used by the time domain smoother. Given that the time domain smoother acts as a control, replacing the unknown diffusion coefficient $c(\cdot)$, we set $k_n = \lfloor 24n^{2/3}/\log(n)^2 \rfloor = 88$, i.e. the ratio of Equation (2.4.1), where $C = 24$, the number of intraday observations. This implies that for each point t , the time domain smoother estimates the volatility using data over the past three and a half days, which is in line with industry standards by arguments in i.a. Guéant (2016) and Fan and Wang (2008) given the hourly interval rate.

5.2 Results

We start with the portfolio consisting of MATIC against XLM and SUSHI against CAKE. Running the test, using the configuration above, showed that there is insufficient evidence to reject the null hypothesis with a test statistic of 3.509 obtaining a p -value of 0.855.

Indeed, if we compare the estimated volatilities obtained by the state domain smoother and the time domain smoother, as shown in Figure 2, we see that both methods are able to capture the true volatility. What is interesting is that, in addition to the conclusion of the stationarity test, Figure 2 also shows that the diffusion coefficient of the process is likely to be time homogeneous. That is to say, when we compare the general shapes of the estimated volatilities and the observed processes, we see that their volatility seems to be state dependent yet not varying as a function of time. Given the time-dependent nature of financial assets, for example due to news influences or macro trends such as the business cycle, a plausible explanation is that any time dependent components present in either MATIC, XLM, SUSHI or CAKE canceled out when expressing one with respect to the other. This would inevitably imply that if either of the processes had a time dependent component not present in the other, then the ergodic hypothesis would conclude that the state and time estimators would be different. Although minor time dependent trends may still be present in the obtained processes, similar to the results of Section 4, it is likely that the sample size is simply too small to obtain sufficiently strong evidence to infer

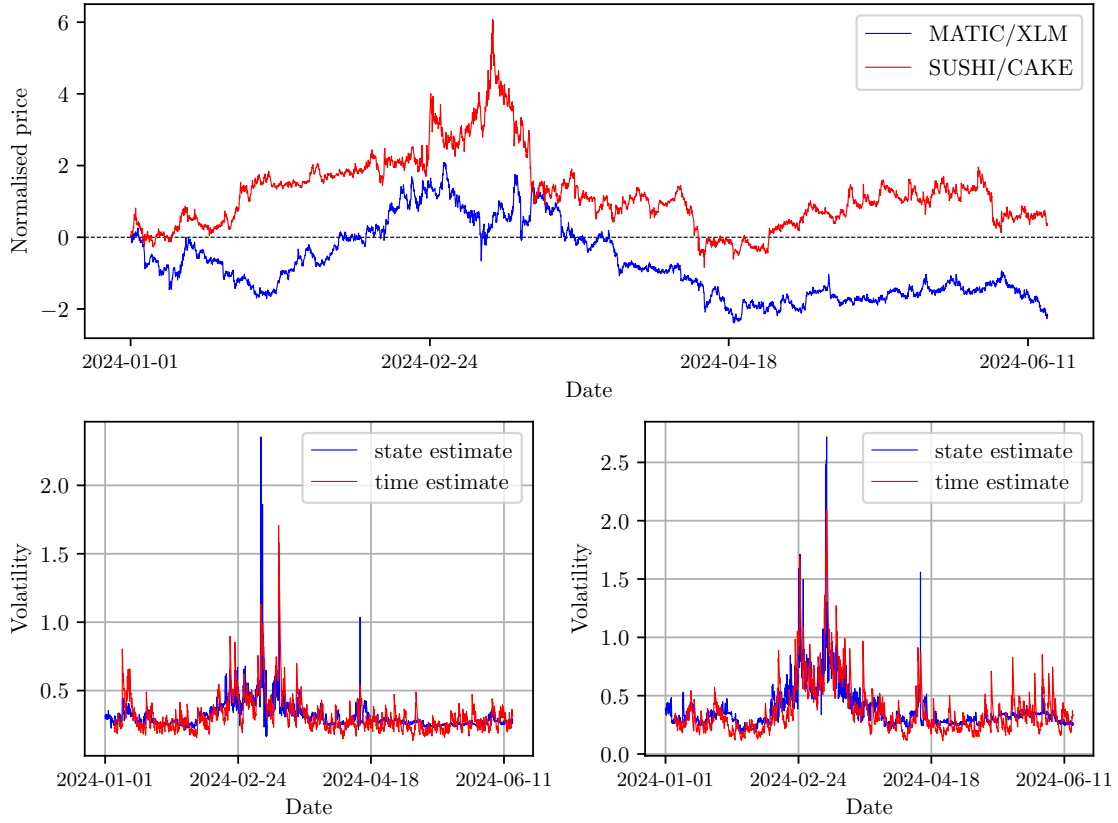


Figure 2: Estimated (daily) volatilities of MATIC against XLM (lower left panel) and SUSHI against CAKE (lower right panel) using the state domain smoother and the time domain smoother.

that this is indeed the case.

Besides the proposed method using the running maximum, it is interesting to additionally see whether alternative methods would result in similar conclusions. Given the plausible stationarity of the process, much like the results obtained by the simulation study, both the Benjamini-Hochberg and Benjamini-Yekutieli procedures do not reject the null of stationarity. Yet, similar to the obtained results of Table 1 to 3, the KPSS and Leybourne-McCabe tests do reject the null of stationarity. As mentioned in Bandi and Corradi (2014), the rejections of the KPSS and Leybourne-McCabe tests are most likely caused by the fact that the Δ -skeleton of the processes are nonlinear Markov chains, especially given the shape of the volatility of Figure 2. However, other possibilities are that the processes resemble near integrated autoregressive processes, which are simply defined to be processes governed by $x_t = \rho x_{t-1} + \sqrt{\Delta_n} \varepsilon_t$ where $|\rho| < 1$ yet ρ is (very) close to one. To statistically distinguish ρ from 1 would require a substantial amount of data, which is often expensive to obtain in financial studies.

As for the second portfolio, composed of the crypto assets Render Token and Fetch.AI, both expressed in Tether, the nonparametric test rejected the null hypothesis of stationarity, achieving a test statistic of 12.51 with a p -value of $2.22 \cdot 10^{-16}$.

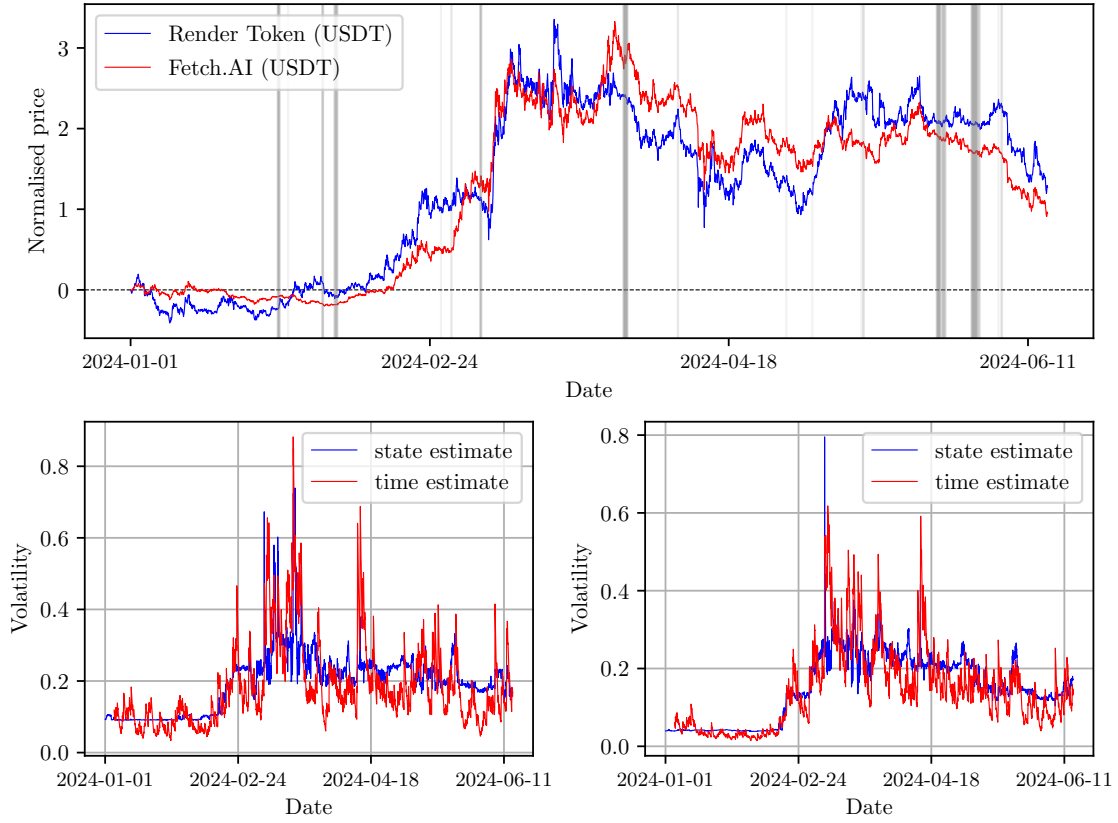


Figure 3: Estimated (daily) volatilities of Render Token expressed in USDT (lower left panel) and Fetch.AI expressed in USDT (lower right panel) using the state domain smoother and the time domain smoother. The upper panel shows the two processes along with the locations where the Gaussian process exceeded the confidence bounds, highlighted by the vertical grey lines.

Given that we constructed the test having a null hypothesis of stationarity, rejection of the null yields the strong conclusion in the sense that there is sufficient evidence to say that the observed process is nonstationary. Contrary to the first portfolio, this time, we can additionally see where the obtained individual test statistics surpassed the confidence bounds associated to the null hypothesis. That is, looking at the grey vertical lines in the upper panel of Figure 3, we see that especially towards the end of the trajectory, there was a discrepancy between the state and time domain smoothing methods (see also Figure 5 for the actual running maximum). Given that the distribution of the data seems to be tight, in the sense of Helly-Prohorov, it is reasonable that the true model is of the time-inhomogeneous type. This is especially clear if we look at the lower two panels in Figure 3, which show the estimated volatilities. Namely, we see that the time domain smoother takes the shape of the ubiquitous generalised autoregressive conditional heteroskedasticity model (or GARCH) while the state domain smoother can poorly adapt to the changing volatility, which would imply that the volatility at a particular state $x \in \mathbb{R}^2$ varies with time. In a similar fashion as Section 4, this then implies rejection of the null hypothesis.

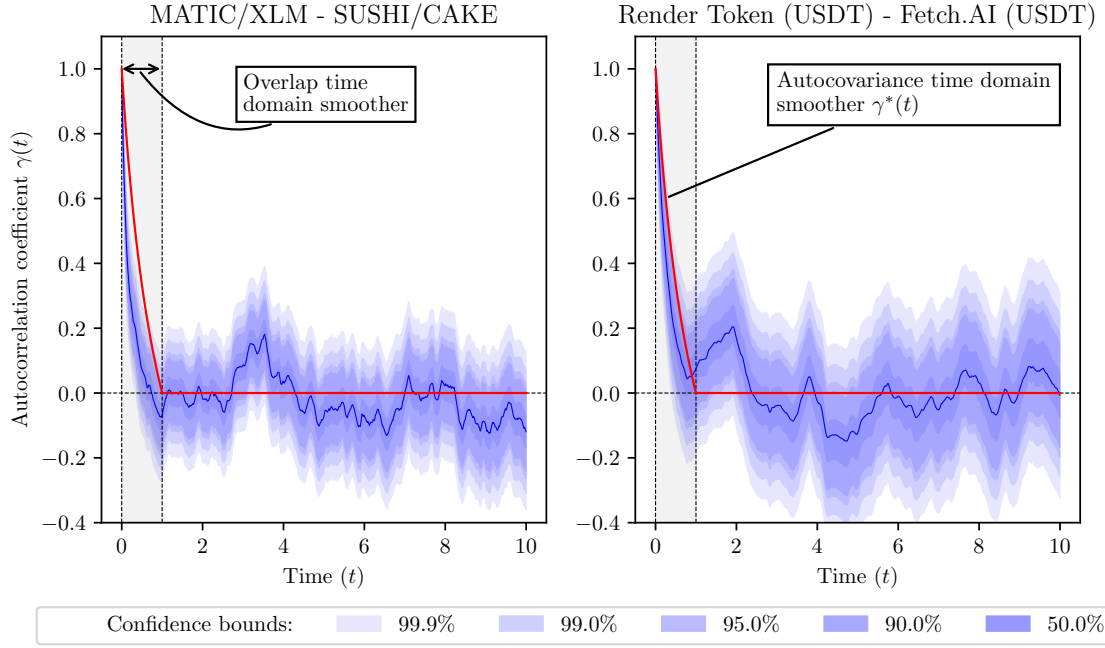


Figure 4: Estimated autocorrelation function on the obtained test statistics using the Bartlett confidence intervals. Both panels also include the interval which has guaranteed overlap in the time domain smoother, showing that this section needs not to have zero autocorrelation. The red marked line represents the autocovariance function of the asymptotic Gaussian process obtained using only the time domain smoother with the index transformation, defined to be $\gamma^*(t) = \frac{1-t \wedge 1}{1+t}$ for $t \geq 0$ (see also Corollary A.2).

As was the case in the simulated environment, both the Benjamini-Hochberg as the Benjamini-Yekutieli procedures rejected the global null hypothesis of stationarity, arriving at the strong conclusion that the portfolio is nonstationary. Similarly, as expected, both the KPSS and Leybourne-McCabe tests additionally arrived at the strong conclusion of nonstationarity. This combined evidence then clearly dictates that indeed, the second portfolio is ineffective for mean-reverting strategies, which in this framework was the underlying motivation to apply the procedure.

Besides the result of the hypothesis itself, it is interesting to analyse whether the assertion of independence in the obtained Gaussian process holds true, and if this dependence satisfies the required convergence rate for increasing yet finite samples. That is, whether the autocorrelation function satisfies $\lim_{t \rightarrow \infty} \log(t)\gamma(t) = 0$ in addition to $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ for $t \rightarrow 0$ with $C > 0$ and $\alpha \in (0, 2]$. Indeed, as discussed in Section 3.1.1, we employed the discrete version of the running maximum distribution obtained by Berman (1982), and so the second condition is not required. However, it would still be interesting to infer the behaviour of γ as if we employed the continuous time version to Z . By the discussion of Section 3, we expect that if $k \leq k_n$, that is the number of data points employed by the time domain smoother, then $|\gamma(k)| > 0$, while $k > k_n$ results in the case where the only source of autocorrelation comes from overlapping data employed by the state domain smoother. Using the

transformation of the index set would imply $|\gamma(t)| > 0$ for $t \in [0, 1)$.

Estimating the autocorrelation function using the Bartlett confidence intervals indeed shows that the obtained transformed Gaussian process satisfies the conditions of [Pickands \(1969\)](#), as can be seen in the left panel of [Figure 4](#). That is, from [Figure 4](#), we see that the autocorrelation rapidly approaches zero, seemingly with an exponential rate. Moreover, compared to the (asymptotic) autocovariance of the time domain smoother, which is simply $\gamma^*(t) = \frac{1-t \wedge 1}{1+t}$ for $t \geq 0$ (c.f., [Corollary A.2](#)), the estimated autocovariance is clearly dominated by γ^* . This essentially implies that any dependence in the state domain smoother vanishes faster than the dependence of the time domain method. Shifting to the properties as $t \rightarrow \infty$, we see that $\lim_{t \rightarrow \infty} \gamma(t) \log(t) = 0$ is likely to be satisfied, as the estimated autocovariance is insignificant from zero at the 95%-confidence level for practically all $t \geq 1$. Interestingly, the estimated autocorrelation function for the nonstationary portfolio, shown in the right panel of [Figure 4](#), additionally seems to satisfy the conditions of [Theorem A.4](#). Although the distribution of the test statistic is unknown, given that the null hypothesis is rejected, the dependence structure of the obtained sequence of test statistics seems to be independent to the truth of the global hypothesis.

At last, it is interesting to also infer the (visual) properties of the obtained sequence of test statistics, shown in [Figure 5](#), by means of their continuity and dependence properties, but more importantly the evolution of the empirically obtained running maximum. To start with the first portfolio, composed of MATIC/XLM and SUSHI/CAKE and shown in the upper panel of [Figure 5](#), we see that the obtained sequence of test statistics seems to be stable. Yet, remarkably, we do find that the mean of the process is 0.39, which is slightly above its theoretical value of zero. One plausible explanation is that the matrix of bias terms $\Gamma(x)$ in [Theorem 2.2](#) is nonzero, given that the results in [Bandi and Moloche \(2018\)](#) only ensure a vanishing influence of the bias correction in an asymptotic sense. Another explanation is that the sample distribution of Z_j^n is negatively skewed, as [Theorem 2.2](#) is a statement on asymptotic normality, meaning that the small sample distribution can be different from its asymptotic limit. Given the estimated shape of the unconditional distribution of the test statistic sequence, obtained using the Gaussian kernel density estimator with Scott's bandwidth rule (see, e.g., [Wasserman \(2006\)](#)), the latter explanation seems logical. Although the obtained empirical distribution does not match the standard normal law, we see that the tail of the obtained distribution behaves as $x \mapsto e^{-x^2/2}$, which is precisely the tail to the standard normal distribution. This essentially implies that the obtained test statistic would be a correct estimate for the running maximum, given that it uses the tail behaviour of Gaussian processes, even though the interquartile range is poorly adjusted to the standard normal law.

However, given the size of the obtained sample, the chosen bandwidth sequence would imply that $\hat{c}_{SD}^n(x)$ is estimated using $nb_{n,T}^2 \approx 296$ observations, which by

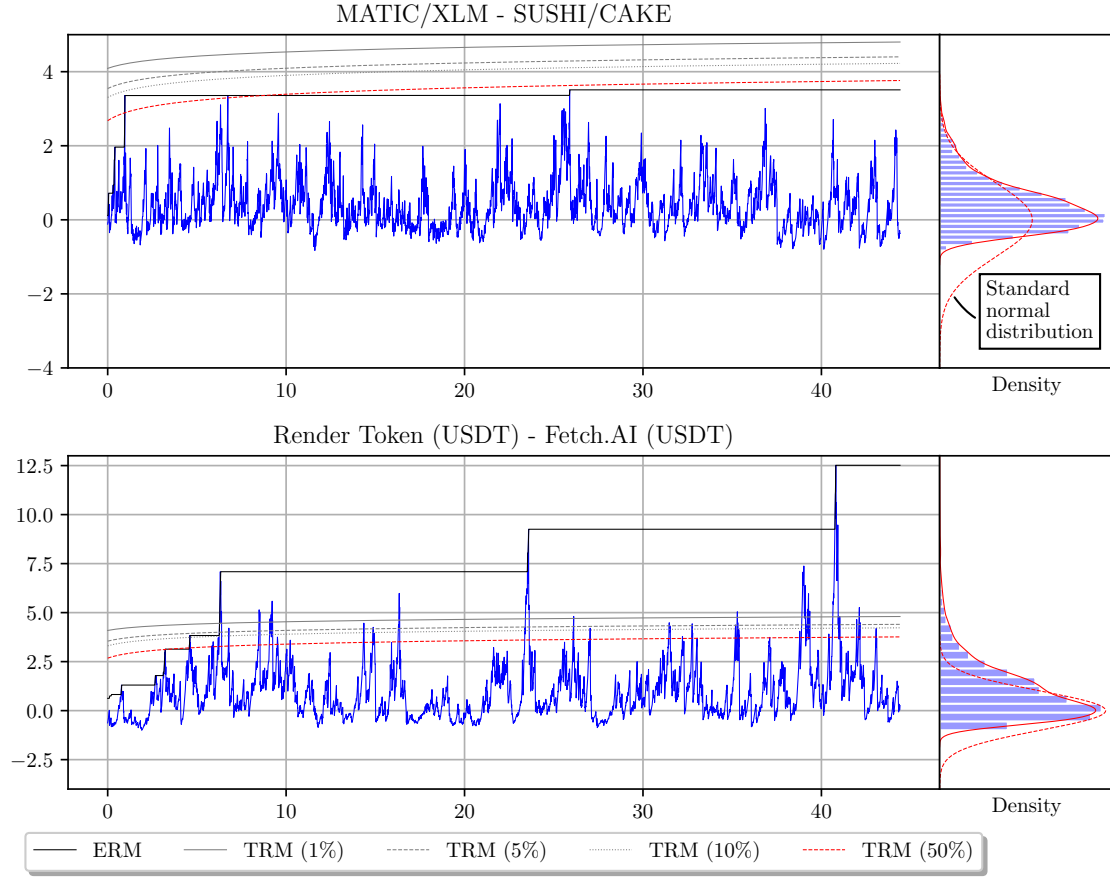


Figure 5: Empirically obtained sequence of test statistics for the MATIC/XLM and SUSHI/CAKE portfolio (upper panel) and the RNDR/USDT and FET/USDT portfolio (lower panel). The black line represents the empirically obtained running maximum (ERM) while the grey lines represent the theoretical running maxima (TRM ($x\%$)) under the associated confidence level obtained using the discrete Pickands critical bounds of [Pickands \(1969\)](#) (c.f., [Proposition A.2](#)). For visual completeness, the 50% theoretical running maximum is additionally included as a dashed red line. Here, the x -axis denotes the transformed times $t_j \mapsto t_j/k_n\Delta_n$ of the index on which X is observed.

[Wasserman \(2006\)](#) should be sufficient to obtain reliable estimates for the asymptotic normal limit. Indeed, for the time domain smoother of [Jacod and Protter \(2011\)](#), the same logic applies, given that $k_n = 88$ should be sufficient to obtain reliable estimates. Given the vanishing dependence as a consequence to [Figure 4](#), the most likely scenario would be that small time dependent components are still present in $c(x) = c(t, x)$, causing the state domain estimates to be slightly higher (on average) than the time domain estimates. These differences are insufficient to reject the global hypothesis of stationarity, but do imply that additional caution is needed when drawing conclusions.

As for the nonstationary portfolio, displayed in the lower panel of [Figure 5](#), we observe that the estimated density is strongly negatively skewed, resembling distributions such as the gamma or F -distribution rather than the normal law. As for the running maximum, the evolution of the extremes quickly surpasses the 95% confi-

dence bounds for the theoretical running maximum, where the obtained sequence exceeds the 95% level at multiple occasions. Here, similar to the first portfolio, the obtained process has a positive mean of 0.97. Yet, we see that the tails of the density are much larger than the expected standard normal tails, likely caused by the sublinear convergence rate of the empirical occupation measure. Given the results of Figure 3, this is expected as the state domain smoother consistently overestimated the volatility, explaining the negative skew of the distribution, from which the extreme tails are obtained as a consequence of scaling with $T/L_{n,T}(x) \rightarrow \infty$ (c.f., Equation (2.4.1)).

6 Conclusion

In this study, we constructed a methodology to test the hypothesis that an observed Itô diffusion process was stationary. Using the work of [Darling and Kac \(1957\)](#), we utilised the discrepancy in convergence rates of the occupation measure between stationary and nonstationary processes. Given their unknown weak properties, we constructed the test statistics using an approach resembling the Durbin-Wu-Hausman test, where we compared the behaviour of two estimators, where only one depends on the convergence of the occupation measure. Using the state domain smoother of [Bandi and Corradi \(2014\)](#) and the time domain smoother of [Jacod and Protter \(2011\)](#), under mild regulatory conditions, we showed that both methods are asymptotically independent and that their standardised difference converges to a standard normal distribution. Consequently, we showed that this essentially results in a sequence of standard normal variables, from which a final test statistic was obtained by means of the distribution of the running maximum introduced in [Berman \(1964, 1982\)](#) and [Pickands \(1969\)](#). Contrary to existing literature, and in particular the results of [Bandi and Corradi \(2014\)](#) and [Kanaya \(2011\)](#), our proposed method is constructed assuming stationarity. Moreover, the proposed method does not suffer from loss of power as was the case in [Bandi and Corradi \(2014\)](#) and [Kanaya \(2011\)](#).

A simulation study, which unveiled the behaviour of the proposed test in the realm of bivariate processes, showed that it had satisfactory size and power. For the simulated stationary Ornstein-Uhlenbeck process, we obtained an empirical rejection rate slightly above 5%, both using the running maximum approach as the solution provided by the Benjamini-Hochberg procedure of [Benjamini and Hochberg \(1995\)](#) although the more sophisticated Benjamini-Yekutieli method of [Benjamini and Yekutieli \(2001\)](#) was too conservative in its decision to reject the global null of stationarity. We also demonstrated that popular linear methods, particularly the KPSS and Leybourne-McCabe tests, become obsolete when applied to continuous time processes, as both methods severely over-reject the null hypothesis of stationarity. When applied to a nonstationary planar Brownian motion, we found that in

smaller samples, the method was unable to reject the majority of simulated sample paths for nonstationarity. However, a mild increase in the sample size corresponded to a rapid increase in observed rejections. Despite this, in terms of power, the running maximum approach did not achieve better results compared to the KPSS and Leybourne-McCabe methods, which are specifically designed for linear models. A practical application on two bivariate financial portfolios aligned with prior placed predictions on the stationary properties of the processes, and an analysis of the dependencies of the obtained test statistics confirmed the theoretical results.

Although the method performed well in the simulated environment, it still has various shortcomings, particularly in obtaining a reliable configuration in small samples. This practical difficulty is induced by the choice to compare two estimator sequences. Circumventing the shortcomings in the method of [Bandi and Corradi \(2014\)](#) came with the trade-off of increased configuration complexity. While hyperparametric tuning can partially mitigate these issues, it inevitably increases estimator uncertainty and can be time-consuming. Additionally, various numerical issues can arise when obtaining the test statistic. For instance, a severely correlated pair of processes can cause difficulties when inverting the asymptotic covariance matrix of the (half-vectorized) difference, potentially rendering the result invalid.

6.1 Future Research

Given the various limitations of the current paper, in this section, we will provide directions and ideas for possible future research. First, we considered to employ the methods of [Bandi and Moloche \(2018\)](#) and [Jacod and Protter \(2011\)](#), yet the test solely relies on the results obtained by [Darling and Kac \(1957\)](#) and so it would be interesting to generalise the procedure to arbitrary methods whose asymptotic properties rely on the divergence rates of the occupation measure. Another generalisation can be achieved by considering a broader class of processes. For example, in recent times, rapid advancements have been made in the estimation of semimartingales (of the Markov type), which by [Jacod and Protter \(2011\)](#) is a much larger class of models.

On a more practical side, a general method to configure the test has yet to surface, which indeed could have detrimental impact on the validity of the test, even if the asymptotic results are viable. Most importantly, both the analysis of the empirical occupation measure as the bandwidth selection require further research to better understand their impact on the conclusions that a test draws. This can in practice be done by means of Monte-Carlo simulation. Using the settings mentioned in [Bandi and Moloche \(2018\)](#), [Puri \(1970\)](#) and [Wasserman \(2006\)](#), one could potentially tabulate optimal bandwidth and occupation measure configurations for a large class of processes, as was for example done to obtain the Dickey-Fuller critical values.

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A Appendix

In this section, and in particular in Appendix A.1, we derive various results concerning the equivalence between stationarity and divergence rates of occupation measures adjusted to our framework. Mainly, we will argue that a necessary and sufficient condition for stationarity is that the divergence rate of occupation measures is strictly linear. Next, in Appendix A.2, we provide various statements regarding the probabilistic aspects to extremes of stationary Gaussian processes. In Appendix A.3, various claims are stated with regards to the convergence of the sequence of test statistics along with the properties of the autocovariance. Lastly, in Appendix A.4, additional simulation results are shown.

A.1 Identification of the Stationary Property

We start this section by demonstrating that the conditions in Assumptions 1 to 3 are sufficient in order to obtain an equivalence between linear divergence in the occupation measure and stationarity of the associated diffusion process. Although the existing literature and results are generally stated in the language of functional analysis (c.f. Taira et al. (2022)), we will depart from this habit by solely focusing on the probabilistic interpretation.

To start, we take an identical construction to the process as sketched by Assumptions 1 to 3. That is, we consider an $\mathbb{R}^d$ -valued Itô diffusion process $X = (X_t)_{t \geq 0}$ serving as the solution

$$X_t = X_0 + \int_0^t b(X_s)ds + \int_0^t \sigma(X_s)dW_s.$$

Given that we do not assume any functional forms on the coefficients b and σ other than the measurability, Lipschitz and linear growth conditions, the stationarity must naturally follow from the properties of the so-called Markov semigroup $P = (P_t)_{t \geq 0}$ of the process. In short, a Markov semigroup is a family of linear operators P_t on the space of bounded continuous functions $C_b(\mathbb{R}^d)$ that satisfy the following properties (Bakry et al. (2014)):

- (i) For every $t \geq 0$, the operator P_t is a self map on $C_b(\mathbb{R}^d)$, that is $P_t : C_b(\mathbb{R}^d) \rightarrow C_b(\mathbb{R}^d)$.
- (ii) $P_0 = \text{Id}$, the identity operator. (initial condition)
- (iii) $P_t \mathbb{1} = \mathbb{1}$, where $\mathbb{1}$ is the constant function equal to 1. (mass conservation)
- (iv) If $f \geq 0$, then $P_t f \geq 0$. (positivity preserving)
- (v) For every $t, s \geq 0$, $P_t P_s = P_{t+s}$. (semigroup property)

Consequently, using the definitions of Bakry et al. (2014), the operator P_t is defined

to be

$$P_t f(x) = \mathbb{E}f(X_t^x) = \mathbb{E}(f(X_t) \mid X_0 = x) \quad \forall x \in \mathbb{R}^d, \quad f \in C_b(\mathbb{R}^d). \quad (\text{A.1.1})$$

Continuing on the semigroup P , as we will soon unveil, the stationarity property is related to the behaviour of the so-called invariant measure μ .

Definition A.1 (Bakry-Gentil-Ledoux (2014)). A measure μ is said to be an invariant measure for the semigroup P if for every bounded continuous function f and every $t \geq 0$

$$\int_{\mathbb{R}^d} P_t f d\mu = \int_{\mathbb{R}^d} f d\mu.$$

That is to say, $\mu P_t f = \mu f$ for every $f \in C_b(\mathbb{R}^d)$.

Although uniqueness of an invariant measure is not guaranteed and requires technical assumptions, in our case Assumption 3 guarantees the existence of a (Borel) invariant measure μ , see Meyn and Tweedie (1993, 2012) for details. Now intuitively, if the invariant measure is in fact a probability measure (or can be normalised up to a probability measure), then by the Markov property, if the process is initialised at μ , then the process will remain in this distribution, leading to strong stationarity. In case that X_0 is initialised by an arbitrary (probability) measure ν , which can be degenerate, then X is merely stationary in the limit. That is, by Meyn and Tweedie (1993) and Lazić and Sandrić (2021), we will obtain convergence in the total variation norm

$$\lim_{t \rightarrow \infty} \|\nu P_t f - \mu f\|_{\text{TV}} = 0 \quad (\text{A.1.2})$$

for every $f \in C_b(\mathbb{R}^d)$. In this case, if μ is a probability measure, then Equation (A.1.2) implies that the process X is ergodic.

Even stronger is that when X is ergodic, then the invariant measure μ is an extreme point in the set of invariant measures under P (c.f. Eisner et al. (2015), Dales (2001) and §7, Ch. 9 in Granas and Dugundji (2003)). This essentially implies that the invariant measure is unique up to multiplicative constants if the process is ergodic. Conversely, if the invariant measure is not finite, then we will still obtain convergence in total variation granted that Assumptions 1 to 3 are met, yet this time to an invariant measure that is not finite, which inevitably implies nonstationarity of the underlying process.

Consequently, identifying whether a process converges to a finite invariant measure will result in the ability to state whether the process is indeed stationary or not. To achieve this, we can invoke the seminal result of Darling and Kac (1957), who demonstrated that the divergence rates of occupation measures is related to the finiteness condition of the underlying invariant measure. Following Geman and Horowitz (1980), we define a set of occupation measures $\lambda = (\lambda_t)_{t \geq 0}$ to be a family of measures satisfying

$$\lambda_t(A) = \int_0^t \mathbb{1}_A(X_s) ds$$

for every $A \in \mathcal{B}(\mathbb{R}^d)$. Here, the interpretation is that the set function λ_t simply measures the amount of time until t that the process X spends in a set A . To link the behaviour of λ to the stationary characteristic of X , we need the following lemma:

Lemma A.1. *Let X be a diffusion process satisfying the conditions in Assumptions 1 to 3. If $\lambda_T(A) \sim T$ as $T \rightarrow \infty$ for every $A \in \mathcal{B}(\mathbb{R}^d)$ with $\mu(A) > 0$, then X is positive Harris recurrent.*

Proof. Using the definition of [Meyn and Tweedie \(1993\)](#), a Markov process X is said to be positive Harris recurrent if its invariant measure μ is finite, i.e., $\mu(\mathbb{R}^d) < \infty$. Hence, if we can show that $\lambda_T(A) \sim T$ as $T \rightarrow \infty$ for every $A \in \mathcal{B}(\mathbb{R}^d)$ of positive measure implies that $\mu < \infty$, then the positive Harris recurrence characteristic follows.

Using the Darling-Kac theorem and the results in [Bandi and Moloche \(2018\)](#) and [Getoor \(1980\)](#), linear divergence of the occupation measure is equivalent to

$$\lim_{T \rightarrow \infty} \frac{\lambda_T(A)}{T} = \lim_{T \rightarrow \infty} T^{-1} \int_0^T \mathbb{1}_A(X_t) dt = p(A)$$

in probability for every $A \in \mathcal{B}(\mathbb{R}^d)$. In fact, the Azema-Kaplan-Duflo-Revuz theorem (c.f., [Revuz and Yor \(1999\)](#)) strengthens this result to almost sure convergence. Now by [Revuz and Yor \(1999\)](#), in this case, the constant functions are μ -integrable, which must inevitably mean that $\mu < \infty$ on $\mathbb{R}^d$, from which we conclude that X is positive Harris recurrent. $\square$

Here, as a consequence to theorem 3 in [Bandi and Moloche \(2018\)](#), we may replace λ_T in Lemma A.1 with the empirical occupation measure $L_{n,T}(x)$. Given that the convergence of the state domain smoother is dependent on $L_{n,T}$, it must follow that the same convergence is dependent on whether X is positive Harris recurrent or not. Hence, to test the stationary hypothesis, what is left to show is that, for one, positive Harris recurrence is equivalent to stationarity and, secondly, strict sublinear growth in $L_{n,T}$ implies nonstationarity.

To do so, it is essential to introduce the rigorous definition of an aperiodic process. Using the definitions of [Lazić and Sandrić \(2021\)](#) and [Prohaska \(2021\)](#), a diffusion process X is called aperiodic if it admits an irreducible skeleton chain. More precisely, there exists a $t_0 > 0$ and a σ -finite measure μ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ such that for every set $A \in \mathcal{B}(\mathbb{R}^d)$, if $\mu(A) > 0$, then

$$\sum_{n=0}^{\infty} Q(nt_0, x, A) > 0 \quad \forall x \in \mathbb{R}^d$$

where Q is the Markov kernel defined by the relation $\mu P_t(A) = \int_{\mathbb{R}^d} Q(t, x, A) d\mu(x)$. For a more rigorous definition, we refer to theorem 5.1 in [Meyn and Tweedie \(1993\)](#).

Lemma A.2 (Lazić-Sandrić (2021)). *Let X be an Itô diffusion process such that $c(x) = \sigma(x)\sigma(x)^\top$ satisfies the uniform ellipticity condition. Then X is open-set irreducible and aperiodic.*

From Lemma A.2, we see that the uniform ellipticity condition (c.f. Assumption 2) guarantees that the diffusion process is aperiodic, regardless of the stationary properties of the process. Although this might seem disconnected from the notion of stationarity, it is actually the missing piece in the puzzle to uniquely identify the stationary property by means of the occupation measure. That is, to show that positive Harris recurrence (as a consequence to linear growth) implies ergodicity.

Theorem A.1. *Let X be a positive Harris recurrent diffusion process satisfying the conditions in Assumption 1 to 3, then the process is ergodic.*

Proof. We know that an ergodic process has a semigroup converging in total variation norm to its unique invariant probability measure, and so all that needs to be done is to show that this condition will be satisfied. To do so, we will make use of the fact that the map $t \mapsto \|\nu P_t f - \mu f\|_{\text{TV}}$ is nonincreasing and Borel measurable for any $f \in C_b(\mathbb{R}^d)$ (c.f. Meyn and Tweedie (2012)). As a consequence, let $P^* = (P_n)_{n \geq 1}$ be a subset of the semigroup P indexed by the natural numbers such that $\|\nu P_n f - \mu f\|_{\text{TV}} \rightarrow 0$ if and only if $\|\nu P_t f - \mu f\|_{\text{TV}} \rightarrow 0$. Now by theorem 10.4.5 in Meyn and Tweedie (2012), as the process $X^* = (X_n^*)_{n \in \mathbb{N}}$ associated to P^* is aperiodic, there exists a further subsequence $P' \subset P^*$ for which the induced process $X' = (X'_{n_k})_{k \geq 1}$ is strongly aperiodic. Consequently, by Meyn and Tweedie (1993), there exists a Harris set $A \subset \mathbb{R}^d$ so that $Q(x) = P_x(X'_{n_k} \in A, k \in \mathbb{N})$ is harmonic for the Δ -skeleton X' . Hence, the function

$$\ell_a(x) = a^{-1} \int_0^a P_t Q(x) dt, \quad P_t \ell_a = \ell_a, \quad t \geq 0$$

is harmonic. As X is recurrent, ℓ is constant by Meyn and Tweedie (1993), and so standard limit theorems for Harris recurrent Markov chains (c.f. theorem 13.3.1 in Meyn and Tweedie (2012)) dictate that $\|P'_n f - \mu f\|_{\text{TV}} \rightarrow 0$ as $n \rightarrow \infty$. Monotonicity of the map $t \mapsto \|P_t f - \mu f\|_{\text{TV}}$ yields the desired result. $\square$

In essence, we have shown that a positive Harris recurrent process is necessarily ergodic under the restriction of uniform ellipticity. The technical reason as to why this assumption is required is to ensure that the topology induced by the total variation norm is sufficiently regular in order for the convergence to take place. Namely, if we drop Assumption 2, then there exists processes for which the semigroup does not converge in total variation norm, see for example Remarks 3 and 4, and Lazić and Sandrić (2021) or Sandrić (2017).

Next, we will show that sublinear growth of the occupation measure must naturally imply nonstationarity, where sublinear growth refers to $\lim_{T \rightarrow \infty} \lambda_T(A)/T = 0$ for

every compact A . Given that stationarity was a byproduct of ergodicity, this entails that we have to show that sublinear growth can only occur in processes that are not ergodic.

Proposition A.1. *Let X be a diffusion process satisfying Assumptions 1 to 3. Then X is nonstationary if $t \mapsto \lambda_t(A)$ is strictly sublinear for every bounded $A \in \mathcal{B}(\mathbb{R}^d)$.*

Proof. From theorem 8.10 in [Eisner et al. \(2015\)](#), a process X is ergodic if and only if the dimension of the fixed point space to P , denoted $\text{Fix } P = \{f \in C_b(\mathbb{R}^d) : Pf = f\}$, is one. Now clearly, if μ is a probability measure, then constant functions are in $L^1(\mu)$. Yet, if μ is not finite, then constant functions are not μ -integrable (see for example proposition 3.10 in [Revuz and Yor \(1999\)](#) and Lemma A.4 with $\gamma(x) = c > 0$). But then, by the quotient-limit theorem (c.f. [Bandi and Moloche \(2018\)](#)), the sublinear growth implies that μ -integrable functions are nonlinear, showing that the dimension of $\text{Fix } P$ is larger than one, or 1 is not a simple eigenvalue of $\text{Fix } P$. Hence, the process is not ergodic, and so nonstationary. $\square$

To close off, using Lemma A.1, Theorem A.1 and Proposition A.1, we can state the main theorem of this section:

Theorem A.2. *Let X be a diffusion process satisfying the conditions in Assumptions 1 to 3 and assume that $X_0 \sim \mu$. Then X is stationary if and only if $L_{n,T} \sim T$ as $T \rightarrow \infty$.*

Proof. Assume that X is stationary, then the Azema-Kaplan-Duflo-Revuz theorem ensures that constant functions are μ -integrable and so $\lim_{T \rightarrow \infty} T^{-1} \int_0^T \mathbb{1}_A(X_t) dt \rightarrow p(A) < \infty$ for $A \in \mathcal{B}(\mathbb{R}^d)$. The equivalence with the empirical occupation measure is a direct consequence to theorem 3 in [Bandi and Moloche \(2018\)](#). Hence, if $L_{n,T} \sim T$ as $T \rightarrow \infty$, then Lemma A.1 and Theorem A.1 ensure that X is stationary.

Conversely, if $L_{n,T} = o_p(T)$ as $T \rightarrow \infty$, then Proposition A.1 ensures that X is indeed nonstationary, completing the proof. $\square$

A.2 Asymptotics for Extremes of Stationary Gaussian Processes

In this section, we will prove some preliminary lemmas required to show the validity of the asymptotic distribution to the running maximum. To do so, let $\phi(x) = (2\pi)^{1/2} e^{-x^2/2}$, define $\bar{\phi}(x) = \int_{-\infty}^x \phi(z) dz = \phi(-\infty, x]$ and write $[x]$ for the largest integer $y \leq x$.

Lemma A.3. *Let $(\xi_n)_{n \in \mathbb{N}}$ be a sequence of numbers so that $\sup_{n \geq 1} |\xi_n| \leq 1$ and $\lim_{n \rightarrow \infty} \xi_n \log(n) = 0$ and write $(u_n)_{n \in \mathbb{N}}$ for a sequence of positive constant such that*

$\sup_{n \geq 1} n|1 - \bar{\phi}(u_n)| < \infty$. Then

$$\lim_{n \rightarrow \infty} n \sum_{j \leq n} |\xi_j| e^{-u_n^2/(1+|\xi_j|)} = 0.$$

Proof. Let $C > 0$ be so that $\sup_{n \geq 1} n|1 - \bar{\phi}(u_n)| \leq C$ and write $\tilde{u}_n$ for the number so that $\sup_{n \geq 1} n|1 - \bar{\phi}(\tilde{u}_n)| = K$, which is unique as $\bar{\phi} > 0$. But then if $\lim_{n \rightarrow \infty} n \sum_{j \leq n} |\xi_j| e^{-\tilde{u}_n^2/(1+|\xi_j|)} = 0$, we clearly have that it holds true for $(u_n)_{n \geq 1}$ by monotonicity in $u \mapsto n|\bar{\phi}(u)|$ for every $n \geq 1$.

Now by the Hoeffding inequality, $u^2\phi(u)/1 + u^2 \leq 1 - \bar{\phi}(u) \leq \phi(u)/u$ (see lemma 2.2.8 in [van der Vaart and Wellner \(2013\)](#) and theorem 1.3.6 in [Dudley \(2014\)](#)), so that $n\tilde{u}_n^{-1}e^{-\tilde{u}_n^2/2} \rightarrow \sqrt{2\pi}C$ and $\tilde{u}_n/2 \log n \rightarrow 1$. Hence, if $\delta = \sup_{n \geq 1} |\xi_n|$ and $\alpha > 0$ is so that $\alpha < (1 - \delta)/(1 + \delta)$, then $n \sum_{j=1}^{[n^\alpha]} |\xi_j| e^{-u_n^2/(1+|\xi_j|)} \leq n^{1+\alpha} e^{-\tilde{u}_n^2/1+\delta} \leq K n^{1+\alpha-2/1+\delta} \tilde{u}_n^{2/1+\delta} \leq K' n^{1+\alpha-2/1+\delta} \log^{1/1+\delta} n \rightarrow 0$. Now define $\delta_n = \sup_{j \geq n} |\xi_j|$ and observe that $\delta_n \log n \leq \sup_{j \geq n} |\xi_j| j \rightarrow 0$ from which we obtain that $\delta_{[n^\alpha]} \tilde{u}_n^2 = (\delta_{[n^\alpha]} \log[n^\alpha]) \tilde{u}_n^2 \log^{-1}[n^\alpha] \rightarrow 0$, meaning that

$$n \sum_{j=[n^\alpha]+1}^n |\xi_j| e^{-u_n^2/(1+|\xi_j|)} \leq n \delta_{[n^\alpha]} e^{-\tilde{u}_n^2} \sum_{j=[n^\alpha]+1}^n e^{\tilde{u}_n^2 |\xi_j|/1+|\xi_j|} \leq C \delta_{[n^\alpha]} \tilde{u}_n^2 e^{\delta_{[n^\alpha]} \tilde{u}_n^2}$$

which goes to zero as $n \rightarrow \infty$, completing the proof. $\square$

Proposition A.2. Let $(\xi_n)_{n \in \mathbb{N}}$ be a stationary sequence of standard Gaussian variables with kernel $\gamma(n)$ satisfying $\lim_{n \rightarrow \infty} \gamma(n) \log(n) = 0$. Let $a_n = (2 \log n)^{1/2}$ and $b_n = a_n - \log \pi \log(n)/2a_n$ for $n \geq e^2$, then for all $x \in \mathbb{R}$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(a_n (\sup_{j \leq n} |\xi_j| - b_n) \leq x \right) = \exp(e^{-x}).$$

Proof. Let $\xi' = (\xi'_n)_{n \geq 1}$ be an i.i.d. sequence of standard normal variables. Then by Lemma A.3, versions of the Slepian-Fernique-Marcus-Shepp Lemma and Sudakov's Minorisation (c.f. prop. A.2.5 in [van der Vaart and Wellner \(2013\)](#) and chapter 1 in [Dudley \(2014\)](#)), we know that $|\mathbb{P}(\sup_{j \leq n} |\xi'_j| \leq u_n) - \mathbb{P}(\sup_{j \leq n} |\xi_j| \leq u_n)| \leq \frac{2}{\pi} n \sum_{j \leq n} |r_j| e^{-u_n^2/(1+|r_j|)} \rightarrow 0$ as $n \rightarrow \infty$ with $u_n = x/\sqrt{2 \log n} + \sqrt{2 \log n} - \log \pi \log n / 2\sqrt{2 \log n}$ and $x \in \mathbb{R}$ arbitrary. Moreover, by the Fisher-Tippett-Gnedenko theorem and the Berman theorems ([Berman \(1982\)](#)), $\mathbb{P}(a_n(\sup_{k \leq n} |\xi_n| - b_n) \leq \log x) \rightarrow e^{-1/x}$ for every $x > 0$ as $n \rightarrow \infty$ and so the Portmanteau lemma concludes the proof. $\square$

Theorem A.3 (Pickands (1969)). Let $\xi = (\xi_t)_{t \geq 0}$ be a separable weakly stationary centered Gaussian process with unit variance. If $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ as $t \rightarrow 0$ for some $C > 0$ and $\alpha \in (0, 2]$ with $\gamma \in L^2(dt)$ or $\lim_{t \rightarrow \infty} \gamma(t) \log(t) = 0$, then

$$\lim_{T \rightarrow \infty} \mathbb{P} \left(a_T \sup_{t \leq T} |\xi_t| - b_T \leq x \right) = \exp(-e^{-x}) \quad \mathbb{P}\text{-a. s.}$$

where $a_T = (2 \log T)^{1/2}$,

$$b_T = 2 \log(T) - \left(\frac{1}{2} - \frac{1}{\alpha} \right) \log \log(T) + \log \left(\frac{C^{1/\alpha} H_\alpha 2^{(2-\alpha)/2\alpha}}{\sqrt{2\pi}} \right)$$

and

$$H_\alpha = \lim_{T \rightarrow \infty} T^{-1} \int_0^T \mathbb{P} \left(\sup_{u \leq t} \varphi(u) > t \right) e^t dt$$

for $\varphi = (\varphi_t)_{t \geq 0}$ a Gaussian process with mean $-|t|^\alpha$ and $\gamma(s, t) = |t|^\alpha + |s|^\alpha - |t-s|^\alpha$.

Proof. The proof is essentially obtained from theorem 2.7.4 in [Giné and Nickl \(2021\)](#) with the adaptation of proposition 2.7.8 in [Giné and Nickl \(2021\)](#) using $u_T = \sup_{t \leq T} |\xi_t|/a_T - b_T$ for $a_T = (2 \log(T))$ and $b_T/a_T = 2 \log(T) - (\frac{1}{2} - \frac{1}{\alpha}) \log \log(T) + \log \left(\frac{C^{1/\alpha} H_\alpha 2^{(2-\alpha)/2\alpha}}{\sqrt{2\pi}} \right)$. See also [Pickands \(1969\)](#) for the original proof and [Berman \(1964, 1982\)](#) for additional results. $\square$

From Theorem [A.3](#), along with the results of [Kallenberg \(1997\)](#), [Berman \(1964\)](#) and [Berman \(1982\)](#), it is possible to find bounds in which the maxima of a Gaussian process takes values with probability one.

Lemma A.4. *if $\gamma \in L^2(\mathbb{R}, dx)$, then $\lim_{x \rightarrow \infty} \gamma(|x|) = 0$.*

Proof. Assume that there exists a real number $a > 0$ such that $|\gamma|(x) > a$ infinitely often and let $x_0 \geq 0$ be a point such that $|\gamma|(x_0) > a$. Let $\delta > 0$ be arbitrary and let $x \in (x_0 - \delta, x_0 + \delta) = A_\delta$, then by the increment inequality, $|\gamma(x_0) - \gamma(x)|^2 \leq 2 - 2\gamma(x_0 - x) \rightarrow 0$ as $\delta \rightarrow 0$. Hence, we may pick δ so that $x \in (x_0 - \delta, x_0 + \delta)$ implies that $|\gamma|(x) > a/2$. But then $\int_{A_\delta} \gamma^2(x) dx \geq \delta a^2/2 > 0$. But by assumption, we can find a sequence $x = (x_n)_{n \geq 1}$ such that $(x_i - \delta, x_i + \delta) \cap (x_j - \delta, x_j + \delta) = \emptyset$ for every $i \neq j$, meaning that

$$\int \gamma^2(x) dx \geq \sum_{n=1}^{\infty} \int_{x_n - \delta}^{x_n + \delta} \gamma^2(x) dx = \infty.$$

Hence, if $\lim_{t \rightarrow \infty} |\gamma|(t) > 0$, then $\gamma \notin L^2(\mathbb{R}, dx)$. $\square$

Corollary A.1. *Let $\xi = (\xi_t)_{t \geq 0}$ be a separable weakly stationary Gaussian process with unit variance satisfying $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ as $t \rightarrow 0$. If either $\gamma \in L^2(\mathbb{R}, dx)$ or $\lim_{t \rightarrow \infty} t^\lambda \gamma(t) = 0$ for some $\lambda > 0$, then*

$$\lim_{t \rightarrow \infty} \mathbb{P} \left(\frac{1}{\alpha} - \frac{1}{2} \leq \frac{a_t \sup_{s \leq t} |\xi_s| - a_t^2}{\log \log(t)} \leq \frac{1}{\alpha} + \frac{1}{2} \right) = 1 \quad (\text{A.2.1})$$

where $a_t = (2 \log(t))^{1/2}$.

Proof. Let $x(\beta, t) = a_t - \beta \log \log(t)/a_t$, then x is monotonically increasing in the second argument and ξ_t is bounded by $x(\beta, t)$ for sufficiently large t if and only if $\sup_{s \leq t} |\xi_s| \leq x(\beta, t)$. Hence, we may replace $\sup_{s \leq t} |\xi_s|$ by ξ_t in theorem 3.1 of [Pickands \(1969\)](#), and so by Lemma A.4, we have the guarantee that $\inf_{t \in [0, 1]} |t|^{-\alpha} (1 - \gamma^2(t)) > 0$. Hence, the result follows by theorem 3.3 in [Pickands \(1969\)](#). $\square$

Remark 12. From Equation (A.2.1) with $\alpha > 0$, we see that $\sup_{s \leq t} |\xi_s| \sim \sqrt{\log(t)} + \log \log(t) / \sqrt{\log(t)}$ for sufficiently large t (c.f., exercise 2.2.22 in [van der Vaart and Wellner \(2013\)](#)), in addition to satisfying some weak version of the law of iterated logarithm. In fact, it is strongly conjectured that $\liminf_{t \rightarrow \infty} (a_t \sup_{s \leq t} |\xi_s| - a_t^2) / \log \log(t) = \frac{1}{\alpha} - \frac{1}{2}$ and $\limsup_{t \rightarrow \infty} (a_t \sup_{s \leq t} |\xi_s| - a_t^2) / \log \log(t) = \frac{1}{\alpha} + \frac{1}{2}$, both with probability 1, although a formal proof (or counterexample) has not appeared yet, see [Pickands \(1969\)](#) for additional comments.

A.3 Some Probabilistic Results

In this section, we will provide various technical results required in order to construct the Gaussian process in Section 3. First, we define the rudiments of weak stochastic convergence needed in this chapter which will serve as the main pillar in proving the convergence to a Gaussian process. We close off by proving various independence properties of functions of stochastic processes.

Definition A.2 (Kallenberg (1997)). Let (T, d) , (S, ρ) be metric spaces where in addition T is locally compact and let $x, x_1, x_2, \dots$ be càdlàg functions with values S .

- (i) We say that $x_n \rightarrow x$ in the locally uniform topology induced by ρ if and only if $\sup_{t \in K} \rho(x_{t,n}, x_t) \rightarrow 0$ for every compact $K \subset T$.
- (ii) The sequence $x_n \rightarrow x$ in the J1-topology if and only if $\lambda_n \circ x_n \rightarrow x$ and $\lambda_n \rightarrow \iota$ in the locally uniform topology for a set (λ_n) of increasing bijections on $\mathbb{R}_0^+$ with identity ι .

For sake of brevity, write $D_T(S)$ to denote the space of càdlàg functions $x : T \rightarrow S$. Moreover, write Z for a stationary centered Gaussian process with unit variance.

Proposition A.3. *If $n, k_n, T \rightarrow \infty$ so that $\Delta_n \rightarrow 0$, then $Z_n \rightarrow Z$ in $D_{[0, \infty)}(\mathbb{R})$ for the J1-topology.*

Proof. By [Silvestrov \(2008\)](#), $Z^n \rightarrow Z$ in $D_{[0, \infty)}(\mathbb{R})$ for the J1-topology granted that the finite dimensional distributions converge along with tightness. By theorem 12.1 in [Billingsley \(2013\)](#), there exists a metric ρ° on $D_{[0, \infty)}(\mathbb{R})$ equivalent to ρ such that $(D_{[0, \infty)}(\mathbb{R}), \rho^\circ)$ is a separable complete metric space, that is a polish space. Hence, if $P^* = (P^n)_{n \geq 1}$ denotes the sequence of probability measures associated to $(Z^n)_{n \geq 1}$, then P^* is tight if and only if P^* is weakly relatively sequentially compact, see [Dales et al. \(2016\)](#) for further definitions. Now define a metric

$w'_\delta(f, t) = \inf_{(x_i)_{i \in I_t}} \sup_{i \in I_t} \sup_{u, v \in [x_i, x_{i+1}]} |f(u) - f(v)|$ on $D_{[0, t)}(\mathbb{R})$ for I_t a finite partition of $[0, t)$. Using the càdlàg extension to the Arzelà-Ascoli theorem (c.f. §C in [Granas and Dugundji \(2003\)](#)), if $A \subset D_{[0, \infty)}(\mathbb{R})$, then A is relatively compact in J1 if and only if A is uniformly bounded in the sense that

$$\sup_{f \in A} \|f\|_\infty := \sup\{x \in \mathbb{R} : \mu(|f| > x) = 0\} < \infty$$

in addition to the modulus of continuity type function vanishing uniformly over A : $\lim_{\delta \downarrow 0} \sup_{f \in A} w'_\delta(f, t) = 0$ for every $t > 0$.

However, there is an equivalent representation of Skorokhod convergence (c.f. [Kallenberg \(1997\)](#)) which is generally easier to prove. That is, using the Skorokhod representation theorem (c.f. [Jacod and Protter \(2011\)](#)), if $Z^n \rightarrow Z$ for the J1-topology then there exists $Z^{n'}$ and Z' , defined on the same filtered space, such that $Z^{n'} \rightarrow Z'$ $\mathbb{P}$ -a.s. as $n \rightarrow \infty$. Clearly, as any coordinate projection $Z_t^n \rightarrow Z_t$ for the weak topology, the convergence of finite dimensional distributions follows. Given that for every $N \in \mathbb{N}$, the set $P_N^* := (P^n)_{1 \leq n \leq N}$ is trivially tight by the Oxtoby-Ulam theorem, we can simply truncate $P^* = (P^n)_{n \geq N}$ for some finite yet large $N \geq 1$. Consequently, by the Portmanteau lemma, for every ε , we can select $N \geq 1$ such that $\sup_{n \geq N} \sup_{f \in C_b} |P^n f - P f| \leq \varepsilon$ and in particular $\sup_{n \geq N} \sup_{f \in C_b} |f(Z^{n'}) - f(Z')| \leq \varepsilon$ by the Skorokhod representation theorem. Now if $\pi_t : Z \mapsto Z_t$ denotes the evaluation map on $\mathbb{R}_0^+$, then $\pi_t Z^{n'} \rightarrow \pi_t Z'$ for every $t \in \{t \geq 0 : Z'(t^-) = Z'(t)\}$. It is clear that $\{\omega : \omega \mapsto Z(\omega) \in C([0, \infty), \mathbb{R})\}$ contains a set of $\mathbb{P}$ -measure one as a consequence to [Belyaev \(1961\)](#) and so along with the above criterion, we obtain $\pi_t Z^{n'} \rightarrow \pi_t Z'$ in the locally uniform topology for every $t \geq 0$. Using theorems 23.8 and 23.9 in [Kallenberg \(1997\)](#) implies that $(Z^n)_{n \in \mathbb{N}}$ is tight and satisfies the conditions in [Billingsley \(2013\)](#), consequently implying convergence for the J1-topology on $D_{[0, \infty)}(\mathbb{R})$ by means of [Silvestrov \(2008\)](#). □

Next, we will focus on various independence properties of the state and time domain smoothers. In essence, the idea is to argue that the uniform continuity criterion on the vector fields b and the diffusion function c ensures that we can linearly approximate X on small open balls. Consequently, normalising the obtained random variable essentially entails that it is a perturbed translation of the difference of Brownian motions. Given that the asymptotic error vanishes and is dominated by the mesh, the independent increments property of Brownian motions implies the result.

Proposition A.4. *Let X be a stationary diffusion process satisfying the conditions of Assumptions 1 to 3. Define*

$$Y_t^n = \frac{c_t^{-1}}{\Delta_n k_n} \sum_{m=1}^{k_n} \Delta_n^m X_t \Delta_n^m X_t^\top.$$

Then $\text{Cov}(Y_t^n, Y_s^n) \rightarrow 0$ as $k_n, n \rightarrow \infty$ for any two distinct s, t granted that $\Delta_n k_n^2 \rightarrow 0$.

Proof. First and foremost, by the Archimedean property, if $t > s$, then (s, t) is nonvoid, and so as $k_n \Delta_n \rightarrow 0$, it must be that there exists an $N \geq 1$ such that $s < s + 2k_n \Delta_n < t$ whenever $n \geq N$. Now by Assumption 1 and Taylor's theorem, we know that c is differentiable and that $c(x+h) = c(x) + \|h\| \nabla c(x) + \frac{1}{2} \|h\|^2 \nabla^2 c(x)$ for h small. Now using the Euler scheme and the arguments in [Revuz and Yor \(1999\)](#), $\lim_{h \rightarrow 0} (X_t - X_{t-h})/\sqrt{h} = \lim_{h \rightarrow 0} \Delta_h X_t/\sqrt{h} = \sigma(X_t)\eta$ for η a standard normal random variable in $\mathbb{R}^d$. As a consequence, we obtain

$$\sigma(X_{t+h})^{-1} X_{t+h} - \sigma(X_t)^{-1} X_t = (\sigma(X_t) + \|h\| \nabla \sigma(X_t) + \|h\|^2 \nabla^2 \sigma(X_t)/2)^{-1} X_{t+h} - \sigma(X_t)^{-1} X_t$$

and so by the Lipschitz continuity, we can bound $\|h\|^2 \nabla^2 \sigma(x)/\sqrt{h} \leq (C^2 \vee 1) \|h\|^2/\sqrt{h} \rightarrow 0$ and $\|h\| \nabla \sigma(X_t)/\sqrt{h} \rightarrow 0$ as $h \rightarrow 0$. Hence, $\lim_{h \rightarrow 0} \sigma(X_{t+h})^{-1} X_{t+h}/\sqrt{h} - \sigma(X_t)^{-1} X_t/\sqrt{h} = \lim_{h \rightarrow 0} \sigma(X_t)^{-1} \Delta_h X_t/\sqrt{h} = \eta$. But then in particular

$$\begin{aligned} & \frac{1}{\Delta_n k_n} \sum_{m=1}^{k_n} (\sigma(X_{t+m\Delta_n})^{-1} X_{t+m\Delta_n} - \sigma(X_t)^{-1} X_t) \\ & \quad \times (\sigma(X_{t+m\Delta_n})^{-1} X_{t+m\Delta_n} - \sigma(X_t)^{-1} X_t)^\top \\ & = \frac{1}{\Delta_n k_n} c(\xi_t)^{-1} \sum_{m=1}^{k_n} \Delta_n^m X_t \Delta_n^m X_t^\top + \sum_{m=1}^{k_n} o_p(\|m\Delta_n\|) \rightarrow \eta^2 \end{aligned}$$

as $\sum_{m=1}^{k_n} o_p(\|m\Delta_n\|) \leq C k_n^2 \Delta_n \rightarrow 0$ for some $C > 0$. Consequently, by the Lévy independent increments theorem as in [Kallenberg \(1997\)](#), it must be that $\eta = \lim_{h \rightarrow 0} \Delta_h W_t/\sqrt{h}$. Hence, the Fubini-Tonelli theorem and prop. 13.8 in [Fritz and Victoir \(2010\)](#) imply that $\eta = \lim_{h \rightarrow 0} \Delta_h W_t/\sqrt{h} \sim N(0, 1)$, which is independent to $\mathcal{F}_s$ for any $s < t$. Using the Archimedean property again implies that Y_t^n becomes independent to Y_s^n as $n \rightarrow \infty$ for any two distinct $s, t \geq 0$. $\square$

Corollary A.2. Let Y_t^n be as in Proposition A.4 and define $Z_t^n = Y_{t\Delta_n k_n}^n$, then $\lim_{n \rightarrow \infty} \gamma_{Z^n}(t) = \gamma^*(t) = \frac{1-t \wedge 1}{1+t}$ for all $t \geq 0$.

Proof. Observe that Z^n is constructed in such a way that a unit interval over the index corresponds to $\Delta_n k_n$ data points, the exact number employed by Y_t^n . Hence, if $\Delta_{k,p}(x) = \{a : x+k \leq a \leq x+p\}$ with $\Delta_{k,p} = \Delta_{k,p}(0)$, then

$$\lim_{n \rightarrow \infty} \gamma_{Z^n}(t) = \frac{1}{|\Delta_{0,t}|} \int_0^1 \mathbb{1}_{\Delta_{0,1}(t)}(x) dx = \frac{1-t \wedge 1}{1+t}, \quad \forall t \geq 0.$$

The independence for $t \geq 1$ is a direct consequence to Proposition A.4. $\square$

Remark 13. Observe that γ^* constitutes a well defined covariance function as $\gamma(t) = 1 - 2|t| + o(|t|)$ for $t \rightarrow 0$ by the Maclaurin expansion at $t = 0$. Using the Bochner-Herglotz theorem (see, e.g., [Eisner et al. \(2015\)](#)) shows that the inverse Fourier

transform of γ^* is everywhere positive, and hence γ^* is positive definite following [Kallenberg \(1997\)](#). In fact, by [Pickands \(1969\)](#), if $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ as $t \rightarrow 0$, then as long as $0 < \alpha \leq 2$ and $C > 0$, the Bochner-Herglotz theorem ensures that γ is positive definite

Corollary A.3. *Let X satisfy the conditions of Proposition [A.4](#), let $\Xi(x, \alpha_n) = \sum_{m=1}^n \mathbb{1}\{|X_{t_m} - x| \leq \alpha_n\}$ for $\alpha_n \rightarrow 0$ and let*

$$Y^n(x) = \frac{c^{-1}(x)}{\Delta_n \Xi(x, \alpha_n)} \sum_{m=1}^n \Delta_n^m X_t \Delta_n^m X_t^\top \mathbb{1}\{|X_{t_m} - x| \leq \alpha_n\}.$$

Then $\text{Cov}(Y^n(x), Y^n(y)) \rightarrow 0$ as $n \rightarrow \infty$ for any two distinct x, y .

Proof. The proof essentially follows from Proposition [A.4](#) if in addition the probability of overlap vanishes. First, observe that $\mathbb{1}\{|X_t - x| \leq \delta\}$ is nontrivial if $\prod_{j \leq d} \mathbb{1}\{|X_t^j - x^j| \leq \delta\}$ is nontrivial. Hence, we only have to consider a single strip $(x - \delta, x + \delta)$ in a plane in $\mathbb{R}^d$, denoted S^* . That is, let $x, y \in S^*$ be arbitrary distinct points, as S^* is polish by Proposition 1.4.12 in [Dales et al. \(2016\)](#), separable and partially ordered by Sierpiński's theorem, then there exists $x' \in S^*$ so that $x \wedge y < x' < x \vee y$. Given that for a Brownian motion $W = (W_t)_{t \geq 0}$, the statement $\mathbb{P}(|W_t| \geq M) = 2 \int_M^\infty (2\pi t)^{-1/2} e^{-x^2/2t} dx \leq \sqrt{t} e^{-M^2/2t} / 2M$ implies $\mathbb{P}(|W_{t+h}| - W_t| \geq h^p) \leq \sqrt{h} e^{-h^{2p}/h} / 2u^p = e^{-h^{2p-1}} / h^{p-1/2}$ for every $\alpha > 0$. It must be that $P(|W_{t+h}| - W_t| \geq \sqrt{h^{1-\varepsilon}}) \rightarrow 0$ as $h \rightarrow 0$ for every $\varepsilon > 0$.

Now Assumption [3](#) and Lemma [A.2](#) imply that X is aperiodic, hence $\mathbb{P}$ -a.s. not constant. But then, the semimartingale representation theorem (c.f. [Jacod and Protter \(2011\)](#) and [Friz and Victoir \(2010\)](#)) ensures that $X^2 = X_0^2 + M + A$ with A strictly increasing. Given that $A = (A_t)_{t \geq 0}$ is càdlàg (hence $\mathbb{P}$ -a.s. continuous by [Revuz and Yor \(1999\)](#)) and $X \in C[0, \infty)$ with probability one, then $M = (M_t)_{t \geq 0}$ is continuous by prop. 1.36 in [Friz and Victoir \(2010\)](#). Now by the Lipschitz and linear growth conditions and Assumption [2](#), for sufficiently small $h > 0$, there exists linear functions f, g for which $|g'| > 0$ on $C([0, h], \mathbb{R}^d)$ such that $X_t = X_0 + tf(t) + g(t)W_t + o(|h|)$ for $t \in [0, h]$ (c.f. [Kallenberg \(1997\)](#)). But then, given that $A(x) = \{t \in [0, \infty) : W_t(\omega) = x\}$ is contained in a Lebesgue null set for a.e. ω , for any distinct s, t with $|s - t| < h$, it must follow that $\{\omega : X_t(\omega) = X_s(\omega)\}$ is contained in a set of probability zero by means of the above arguments. $\square$

In consequence, using Proposition [A.4](#) and Corollary [A.3](#), we can state the following useful result:

Theorem A.4. *Let X be a stationary diffusion process satisfying Assumptions [1](#) to [3](#), let Z be the Gaussian process from Proposition [A.3](#) obtained from X . If $T \sim n^{(d-1)/(d+4)} \log(n)^{d-1}$ with $k_n = C'n^{4/d+4} / \log(n)^d$, then Z satisfies $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ as $t \rightarrow 0$ where $C > 0$ and $\alpha \in (0, 2]$ with $\lim_{t \rightarrow \infty} \gamma_Z(t) \log(t) = 0$. Moreover, $\{\omega : Z(\omega) \in C[0, \infty)\}$ contains a set of $\mathbb{P}$ -measure one.*

Proof. We start by proving the first assertion. To do so, observe that $b_n/\Delta_n k_n = Cn \log(n)^d / T \log(n) n^{4/d+4} = C''$ for every $d \geq 1$, implying that the state and time domain bandwidth sequences are proportional. Define $\gamma_{0,\beta}^*(t) = \beta \gamma^*(t)/(1 - \beta)$ and $\gamma_{1,\beta}^*(t) = \gamma^*(t) \vee \beta$, then by proportionality and Corollary A.2, we know that $\gamma_{0,\beta}^*(t) \leq \gamma_Z(t) \leq \gamma_{1,\beta}^*(t)$ for $\beta = C''/(1 + C'')$ for sufficiently small $t > 0$ with $\gamma_Z(0) = 1$ trivially. Moreover, as $b_n/\Delta_n k_n = O(1)$ is nontrivially bounded, it must be that Proposition A.4 and Corollary A.3 imply that for some $t' \geq 1$ bounded, $\gamma_Z(|t|) = 0$ for every $|t| \geq t'$. Consequently, as $\gamma(t) = 0$ on $[|t'|, \infty)$, Corollary A.2 implies $\lim_{t \rightarrow \infty} \gamma(t) \log(t)$, meaning that $\gamma \in C_0([0, \infty), \mathbb{R}) \subset L^\infty(\mathbb{R}, dx)$, (this is condition 1.2 in Pickands (1969) as an alternative to $\gamma(t) = o(1/\log(t))$). Using Taylor's theorem, we additionally see that $\gamma^*(t)$ has a Maclaurin expansion at $t = 0$ of $\gamma^*(x) = 1 - 2 \sum_{n=1}^{\infty} (-1)^{n+1} |x|^n = 1 - 2|x| + o(|x|^2)$. Consequently, we see that $\alpha \in (0, 2]$ and $C > 0$. To close off, we note that $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ for $C \in (0, \infty)$ and $\alpha \in (0, 2]$ as $t \rightarrow 0$ is a sufficient condition to ensure that $\omega \mapsto Z(\omega)$ is of class $C[0, \infty)$ $\mathbb{P}$ -a.s., by Belyaev (1961). In fact, the latter result implies that $t \mapsto \gamma_Z(t)$ is continuous by Kallenberg (1997). $\square$

Remark 14. In the book by Giné and Nickl (2021), (a version of) the above result is stated with the additional restriction on $\mathbb{P}$ -a.s. continuity of sample paths, which is redundant as this is implied by $\gamma(t) = 1 - C|t|^\alpha + o(|t|^\alpha)$ as $t \rightarrow 0$ by Belyaev (1961) (see also Pickands (1969)).

A.4 Additional Results

Below contains a tabulated version to the results of Section 4 using a 1% and 10% confidence level.

Bivariate Ornstein-Uhlenbeck					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	1%	1%	1%	1%	1%
50	0.070 (0.0003)	0.150 (0.0006)	0.080 (0.0004)	1.000 (0.000)	0.965 (0.0002)
150	0.015 (0.0001)	0.065 (0.0003)	0.015 (0.0001)	1.000 (0.000)	0.880 (0.0005)
365	0.060 (0.0003)	0.085 (0.0004)	0.065 (0.0003)	1.000 (0.000)	0.760 (0.0009)
Planar Brownian motion					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	1%	1%	1%	1%	1%
50	0.165 (0.0007)	0.355 (0.0011)	0.215 (0.0008)	1.000 (0.000)	0.990 (0.000)
150	0.615 (0.0012)	0.765 (0.0009)	0.645 (0.0011)	1.000 (0.000)	0.990 (0.000)
365	0.900 (0.0004)	0.990 (0.000)	0.915 (0.0004)	0.000 (0.000)	0.000 (0.000)
Bivariate inhomogeneous diffusion					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	1%	1%	1%	1%	1%
50	0.185 (0.0008)	0.410 (0.0012)	0.230 (0.0009)	1.000 (0.000)	0.935 (0.0003)
150	0.120 (0.0005)	0.280 (0.001)	0.155 (0.0007)	1.000 (0.000)	0.865 (0.0006)
365	0.135 (0.0006)	0.195 (0.0008)	0.140 (0.0006)	1.000 (0.000)	0.695 (0.0011)

Table 2: Rejection rates of the Euler-Maruyama simulated diffusion processes with the Milstein correction for $T \in \{50, 150, 365\}$, $n \in \{1000, 3000, 7300\}$ and $\Delta_n = 1/20$ using a version of the Fisher-Tippett-Gnedenko running maximum theorem and the Benjamini-Hochberg and Benjamini-Yekutieli procedures.

Bivariate Ornstein-Uhlenbeck					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	10%	10%	10%	10%	10%
50	0.125 (0.0005)	0.080 (0.0004)	0.045 (0.0002)	0.425 (0.0012)	0.955 (0.0002)
150	0.065 (0.0003)	0.015 (0.0001)	0.015 (0.0001)	0.340 (0.0011)	0.835 (0.0007)
365	0.085 (0.0004)	0.065 (0.0003)	0.040 (0.0002)	0.210 (0.0008)	0.705 (0.001)
Planar Brownian motion					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	10%	10%	10%	10%	10%
50	0.335 (0.0011)	0.195 (0.0008)	0.120 (0.0005)	0.825 (0.0007)	0.990 (0.000)
150	0.765 (0.0009)	0.640 (0.0012)	0.535 (0.0012)	0.930 (0.0003)	0.990 (0.000)
365	0.985 (0.0001)	0.915 (0.0004)	0.840 (0.0007)	1.000 (0.000)	1.000 (0.000)
Bivariate inhomogeneous diffusion					
	Running Maximum	Benjamini-Hochberg	Benjamini-Yekutieli	KPSS	Leybourne-McCabe
T	10%	10%	10%	10%	10%
50	0.375 (0.0012)	0.210 (0.0008)	0.105 (0.0005)	0.395 (0.0012)	0.900 (0.0004)
150	0.275 (0.001)	0.155 (0.0007)	0.090 (0.0004)	0.280 (0.001)	0.810 (0.0008)
365	0.195 (0.0008)	0.140 (0.0006)	0.100 (0.0004)	0.190 (0.0008)	0.640 (0.0012)

Table 3: Rejection rates of the Euler-Maruyama simulated diffusion processes with the Milstein correction for $T \in \{50, 150, 365\}$, $n \in \{1000, 3000, 7300\}$ and $\Delta_n = 1/20$ using a version of the Fisher-Tippett-Gnedenko running maximum theorem and the Benjamini-Hochberg and Benjamini-Yekutieli procedures.